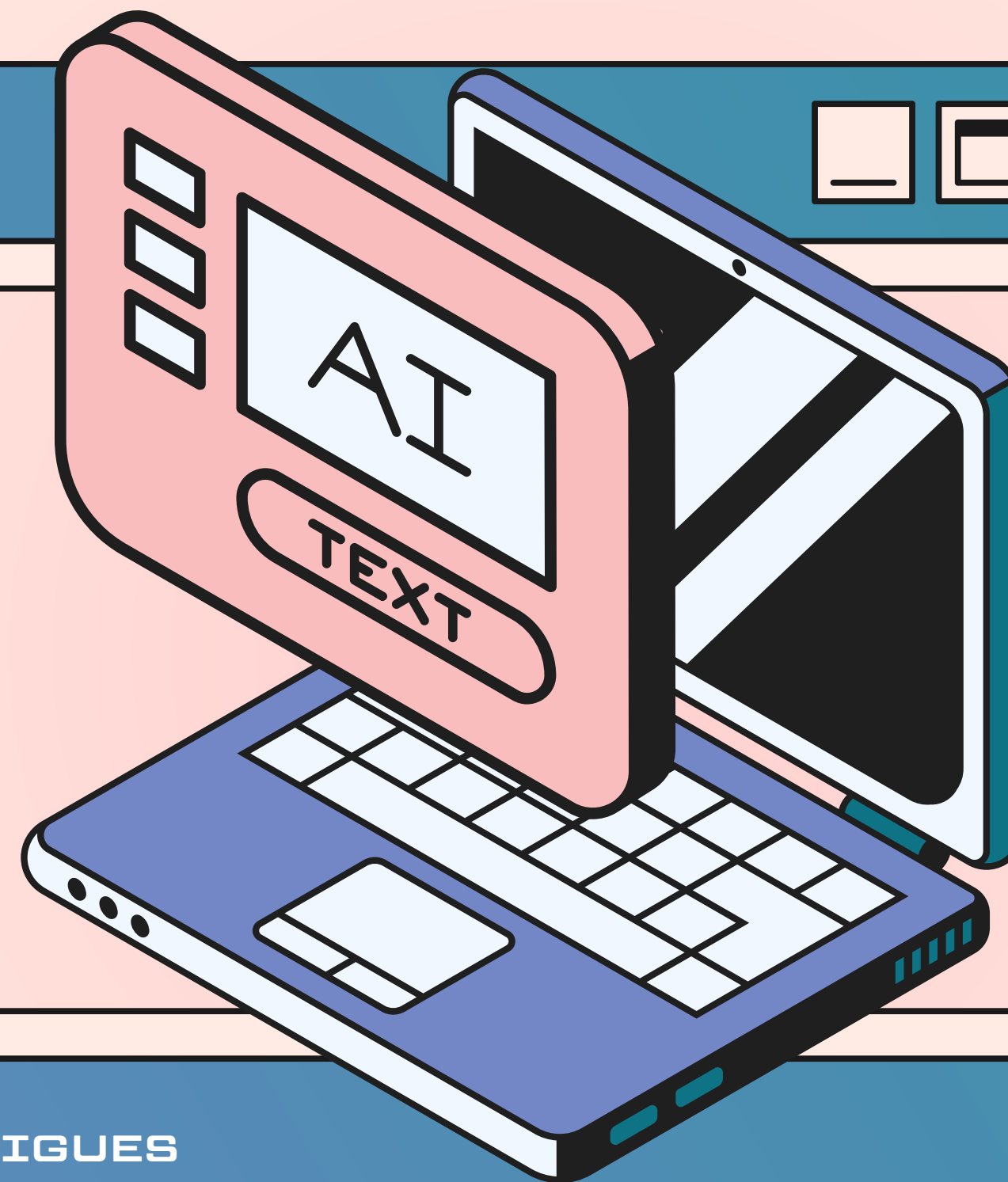




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LISTA DE EXERCÍCIOS

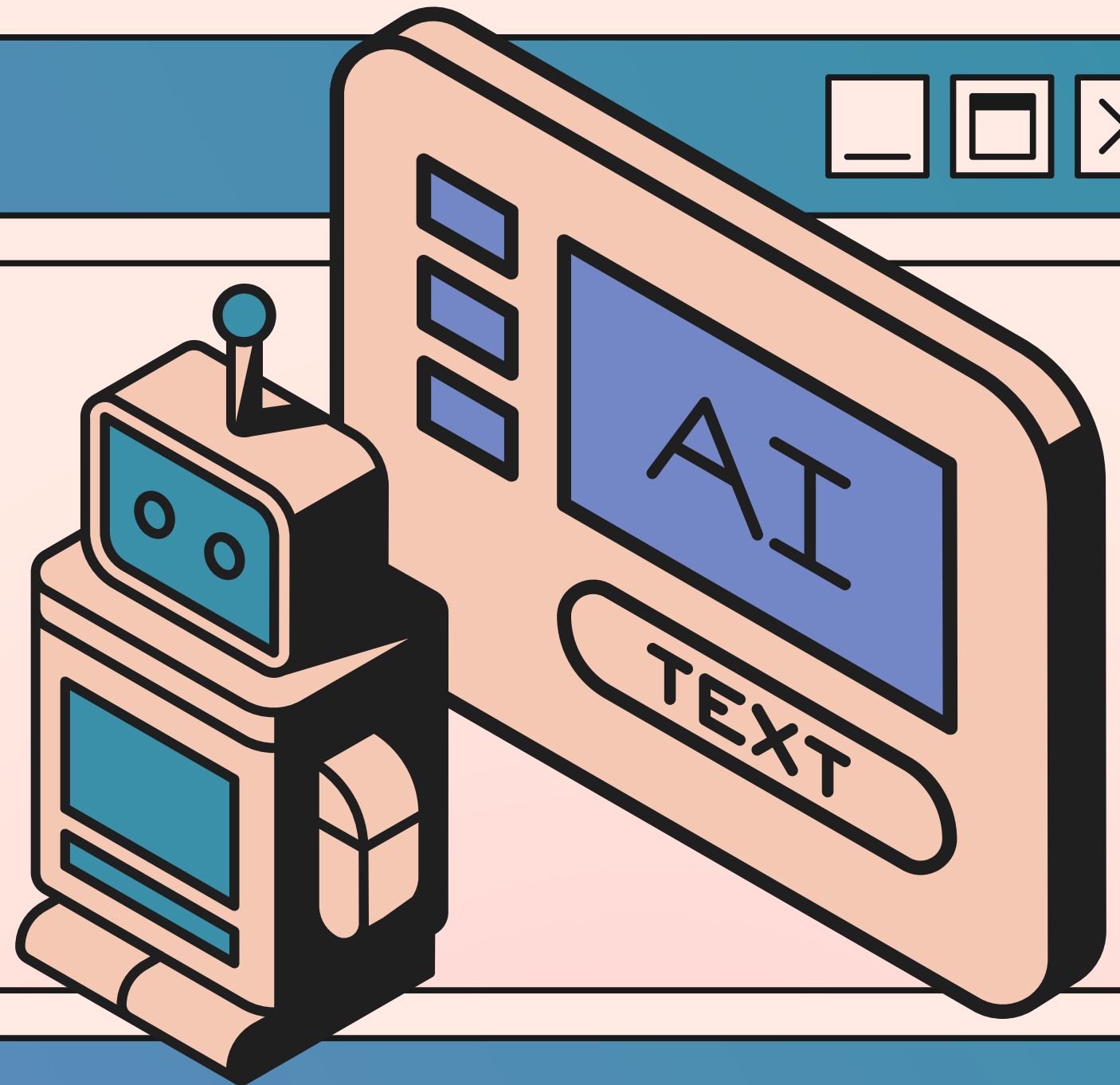


GISELE MENEZES , MARIANA MARTINS E THIAGO RODRIGUES



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DEEP LEARNING - 2025.1

QUESTÃO 1



GISELE MENEZES , MARIANA MARTINS E THIAGO RODRIGUES

QUESTÃO 1

- Criou-se amostras com dados aleatórios a partir das funções dadas.
- 1024 amostras para treino. 256 para validação e teste.
- Amostras foram transformadas em tensores.
- Utilizou-se **Pytorch** para criar o modelo.
- No treino houve a normalização.

Função a:

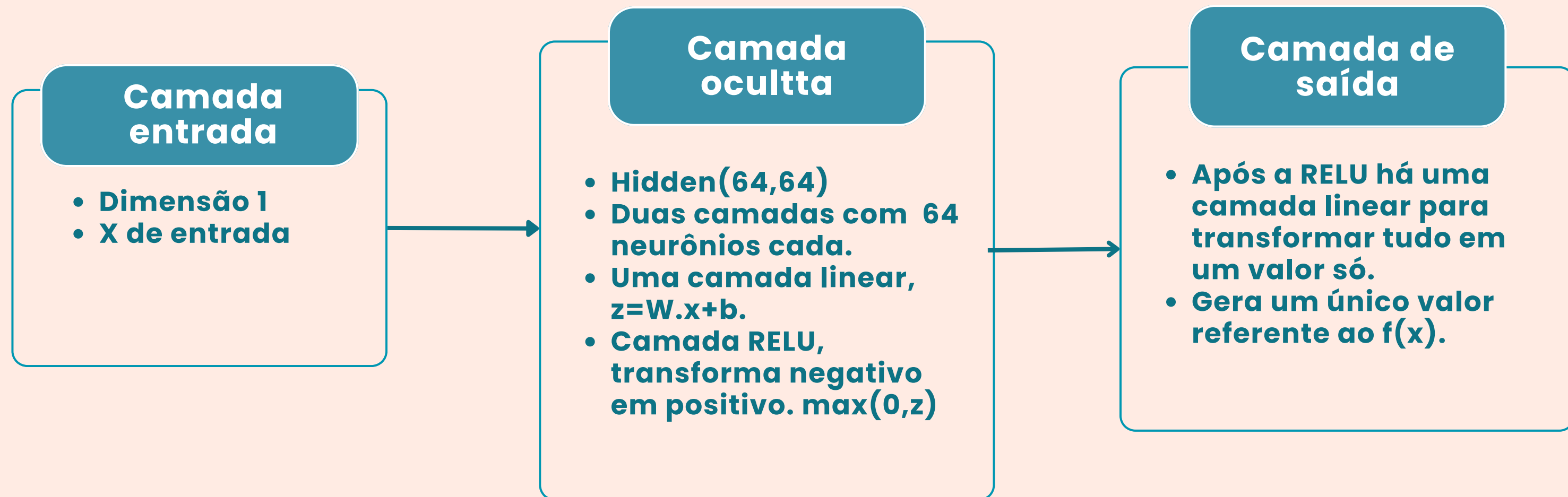
```
x_train shape: (1024, 1) | y_train shape: (1024, 1)
x_validation shape: (256, 1) | y_validation shape: (256, 1)
x_test shape: (256, 1) | y_test shape: (256, 1)
x_mean/std: [2.494779] [1.4463867]
y_mean/std: [0.23734441] [0.87528676]
```

Função b :

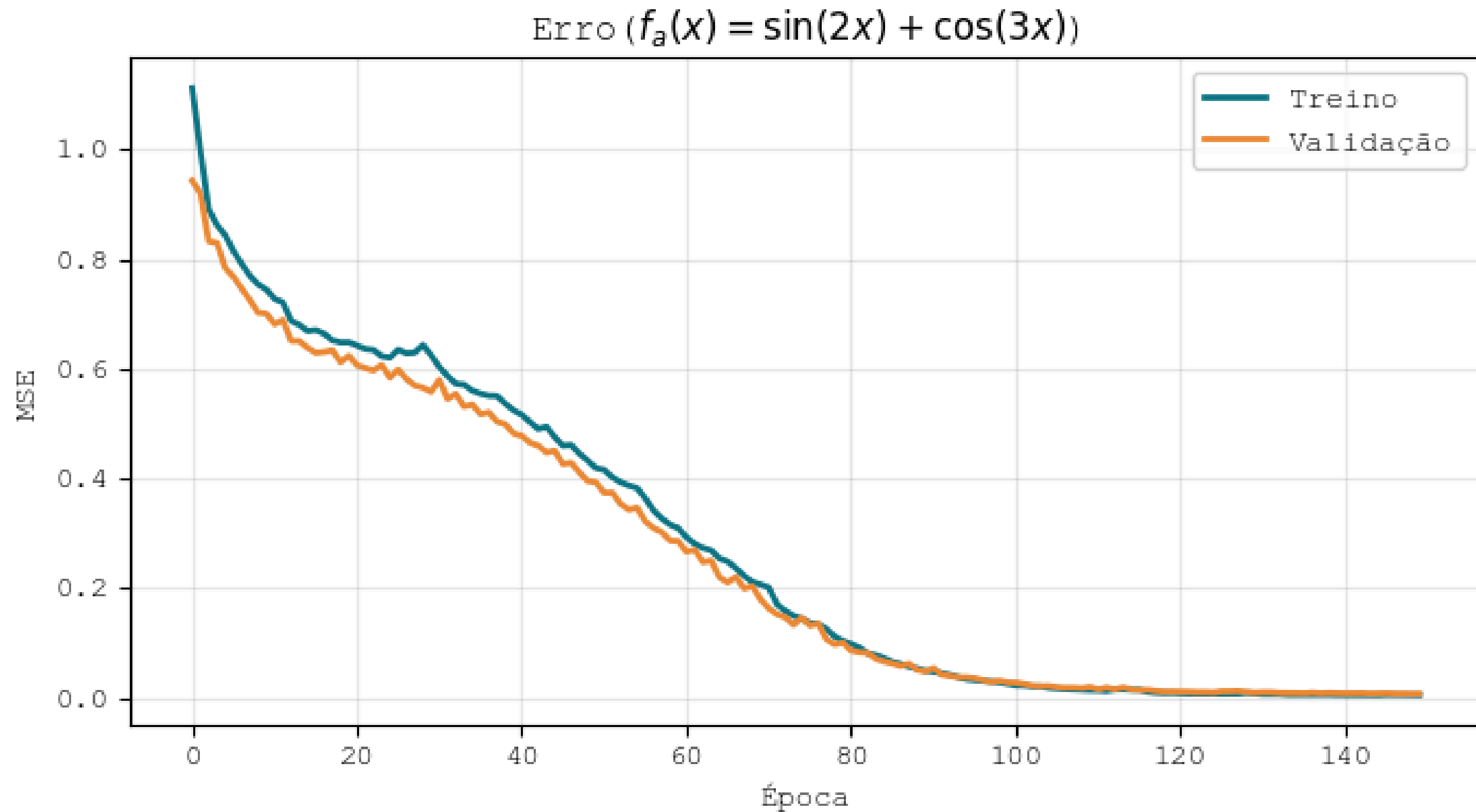
```
x_train shape: (1024, 1) | y_train shape: (1024, 1)
x_validation shape: (256, 1) | y_validation shape: (256, 1)
x_test shape: (256, 1) | y_test shape: (256, 1)
x_validation shape: (256, 1) | y_validation shape: (256, 1)
x_mean/std: [2.5395248] [1.4613246]
y_mean/std: [6239.597] [8975.806]
```

ARQUITETURA

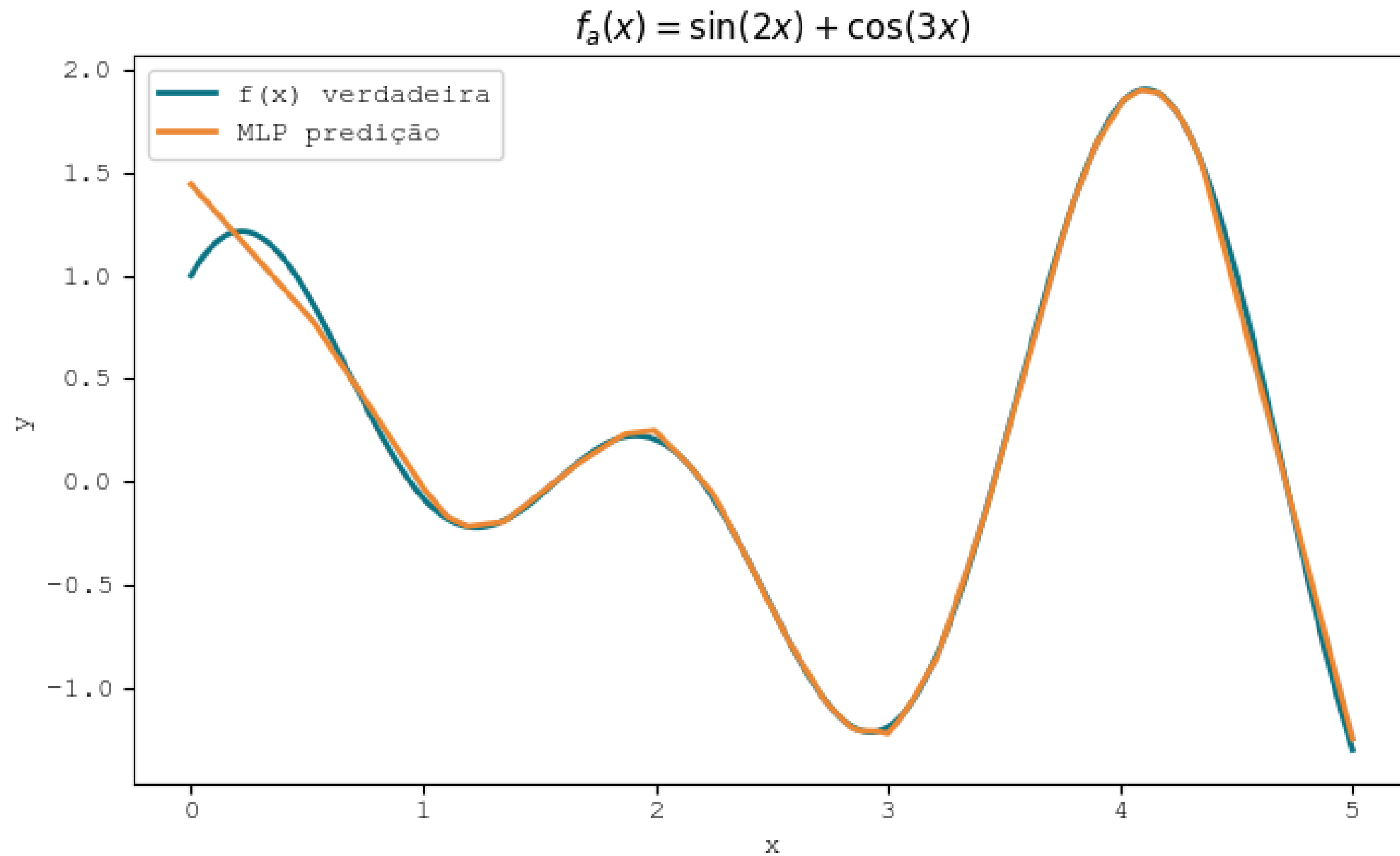
Multi-Layer Perceptron



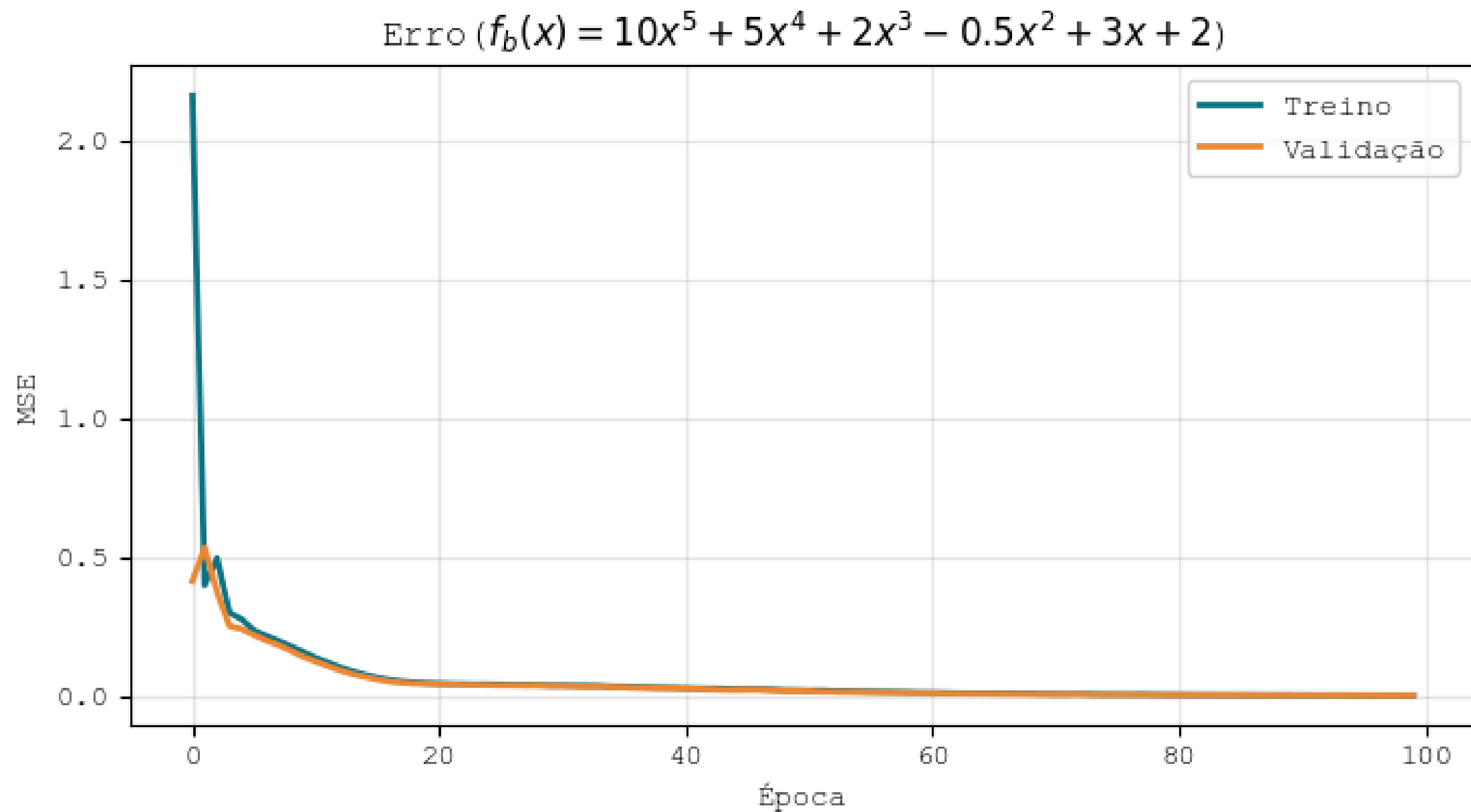
QUESTÃO 1 - A



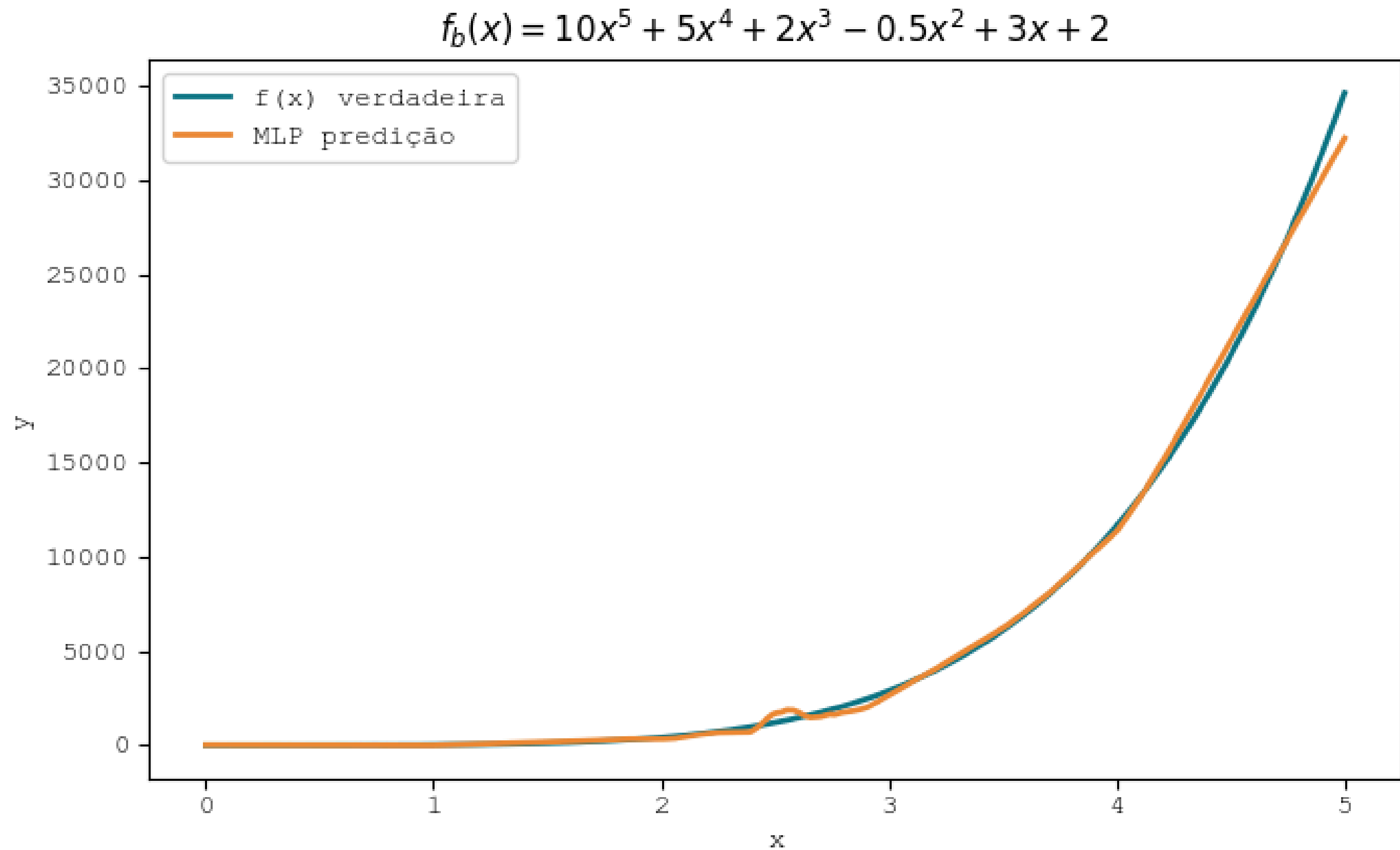
QUESTÃO 1 - A



QUESTÃO 1 - B



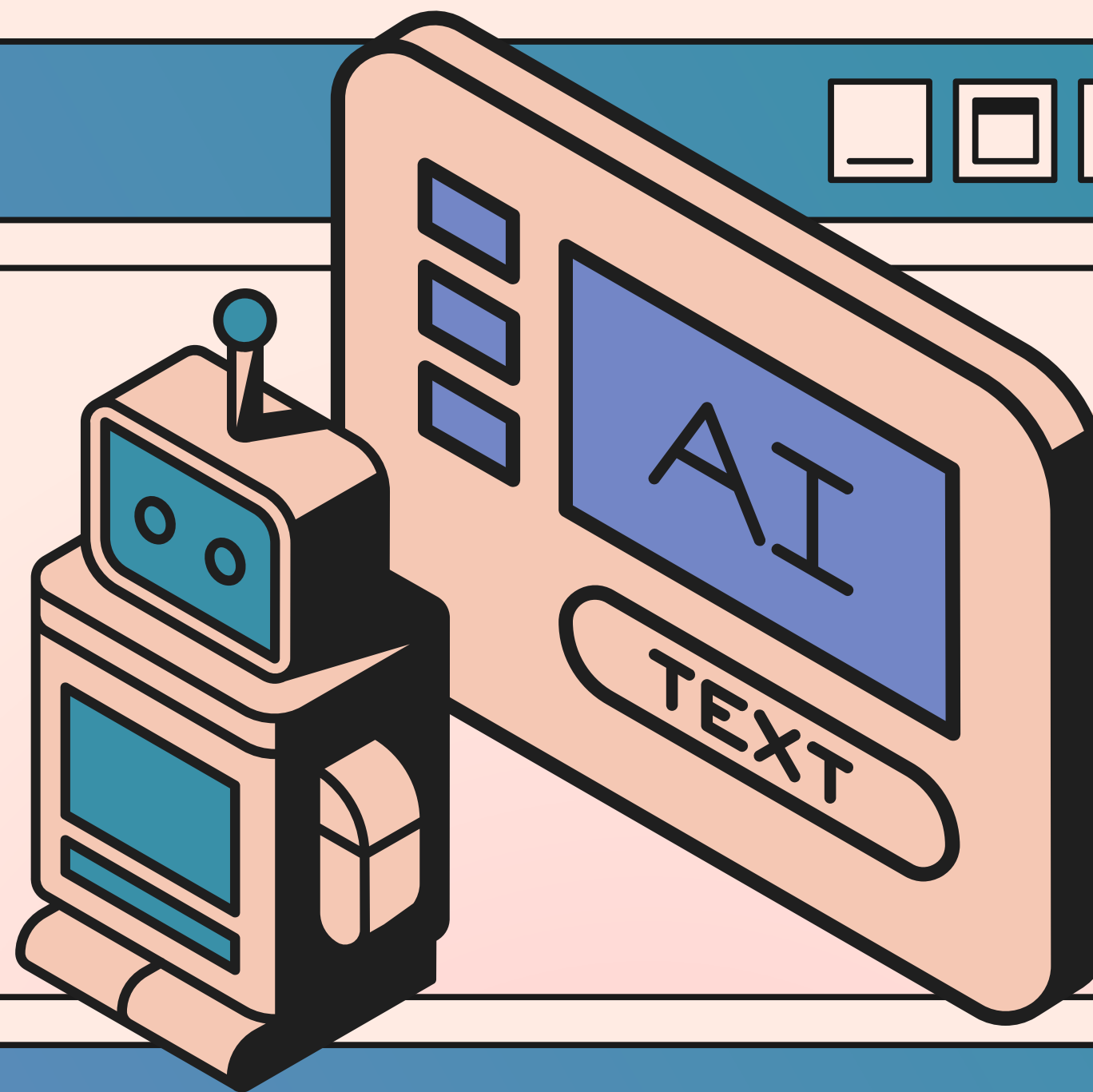
QUESTÃO 1 - B





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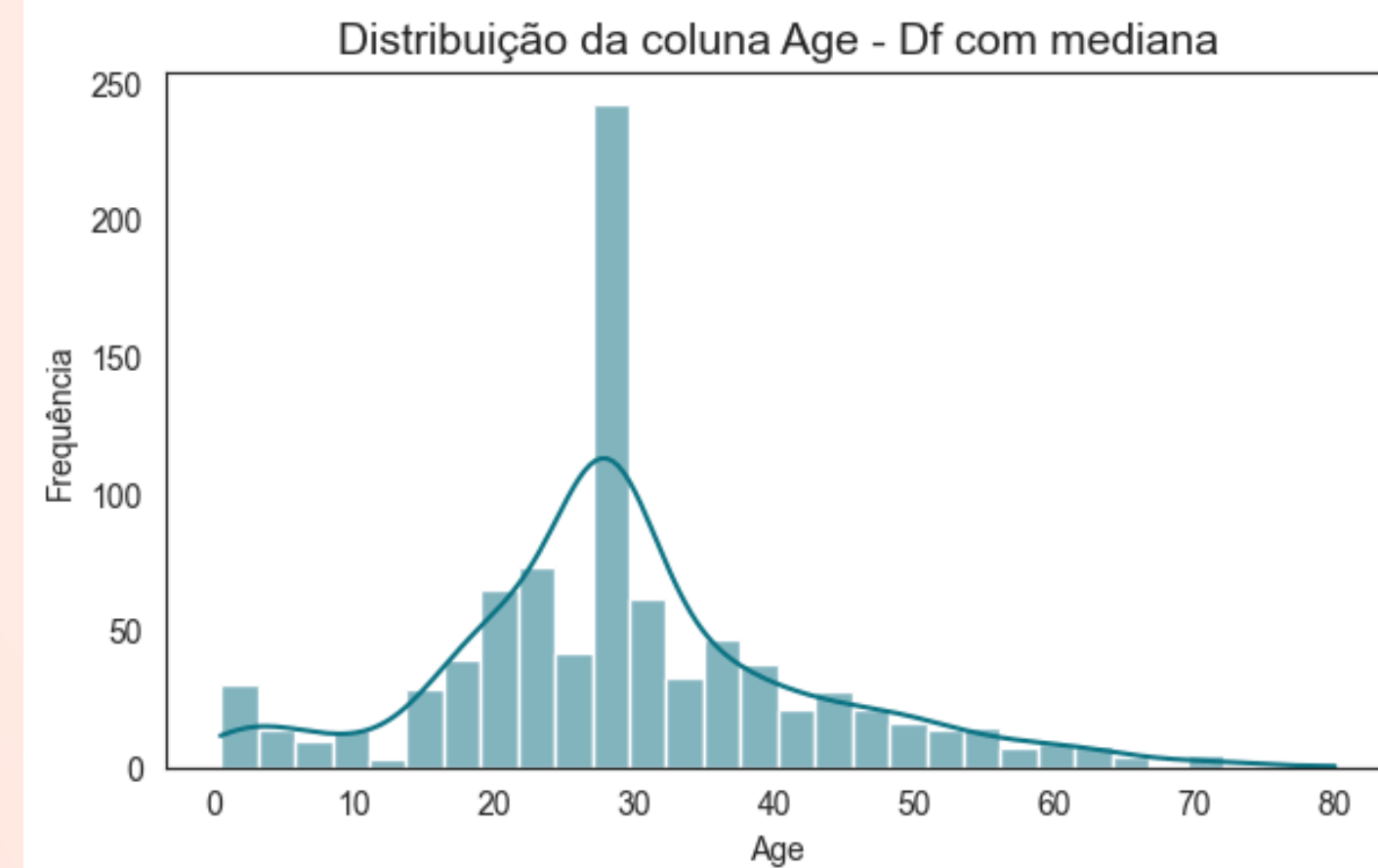
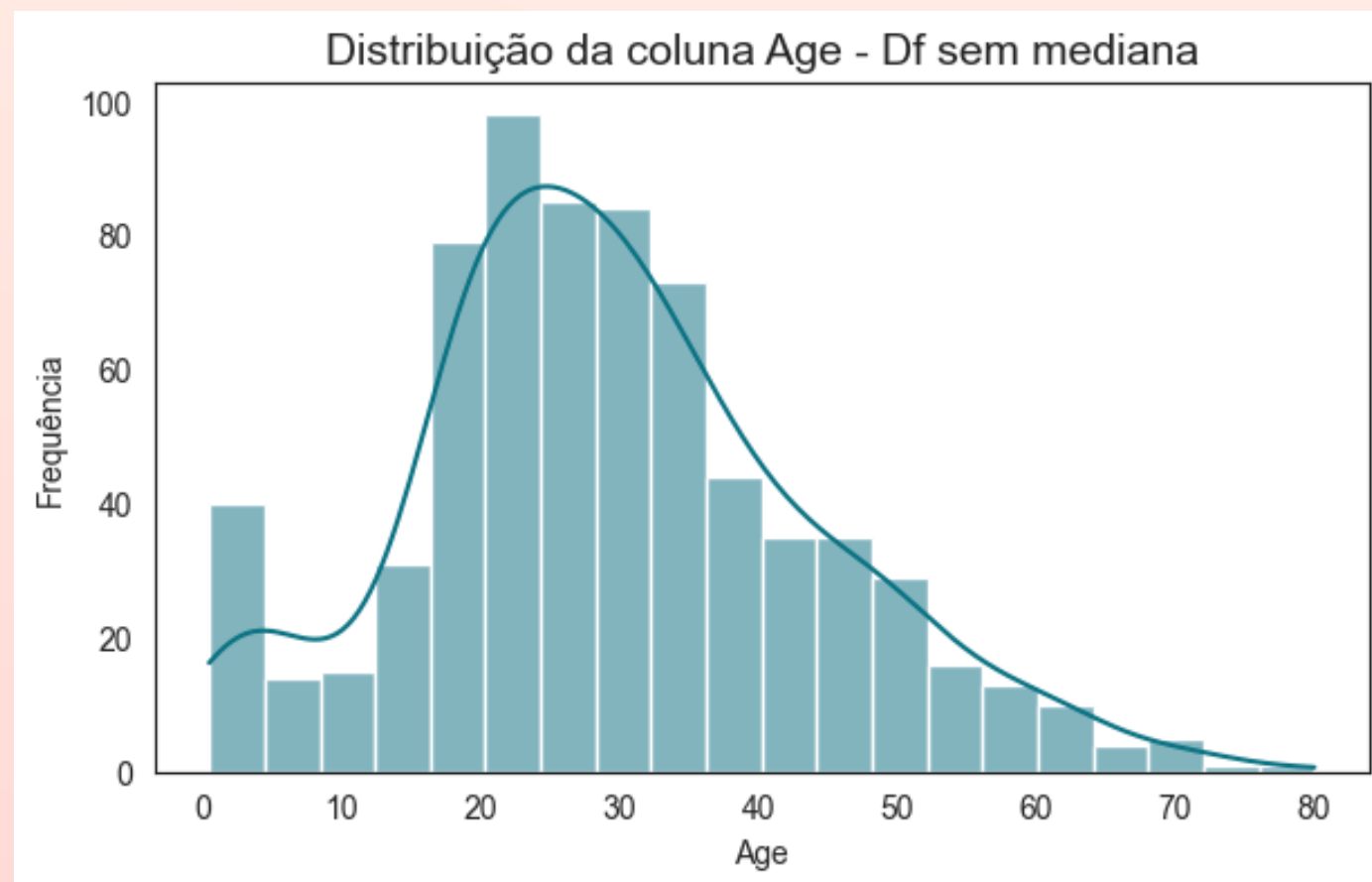
QUESTÃO 2



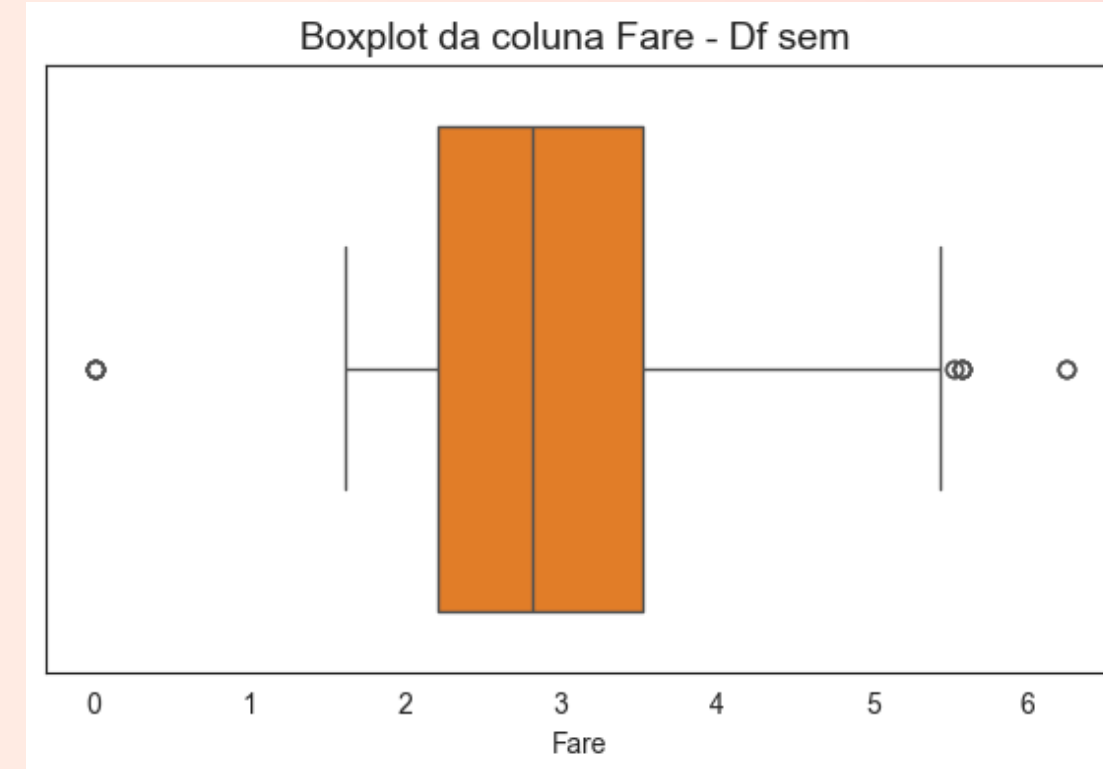
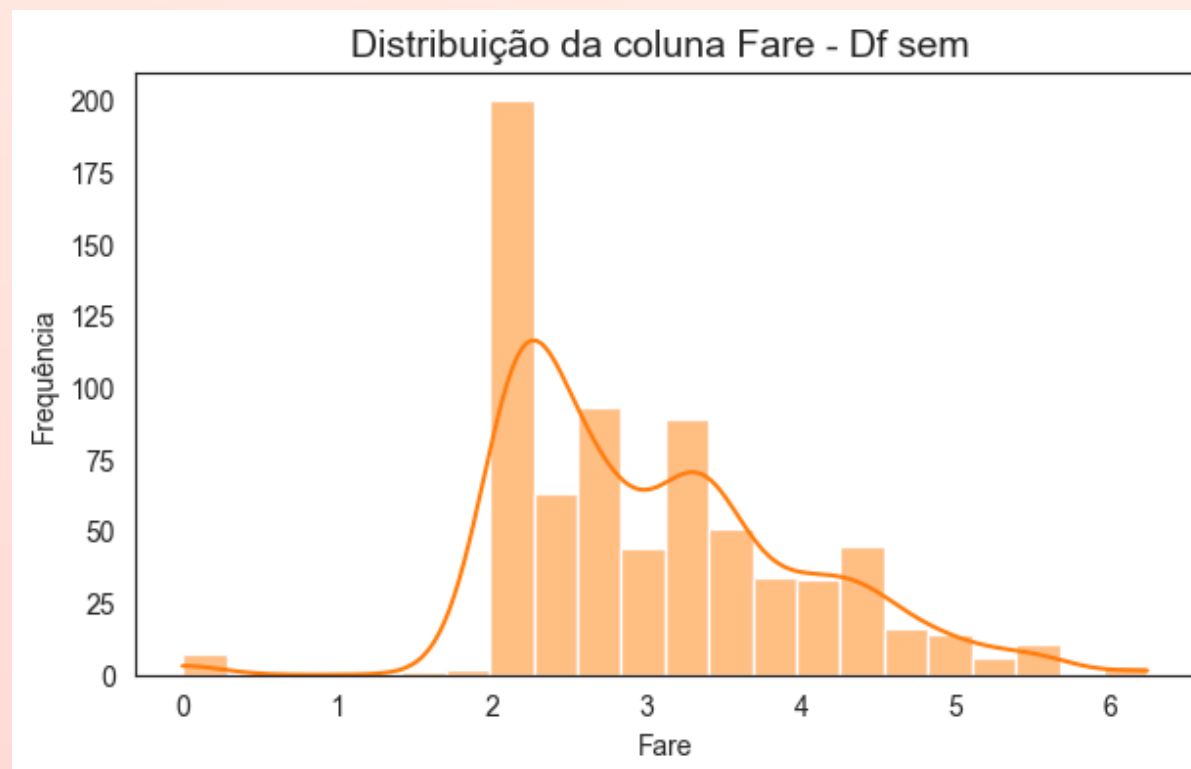
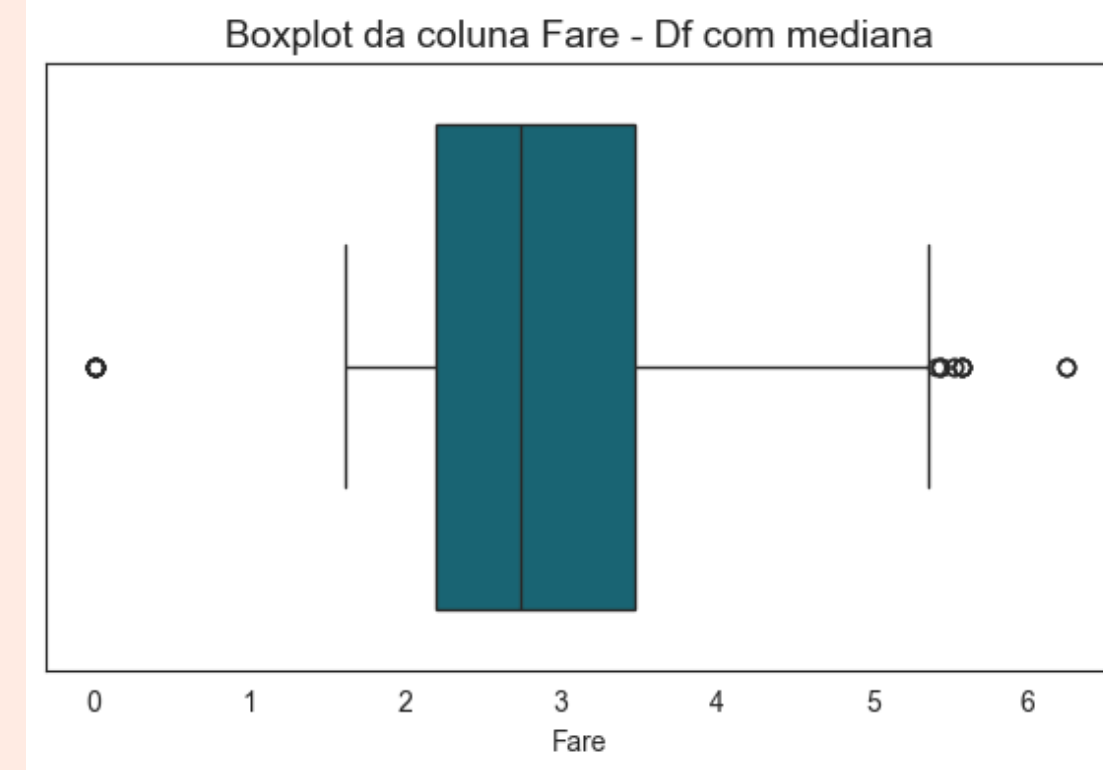
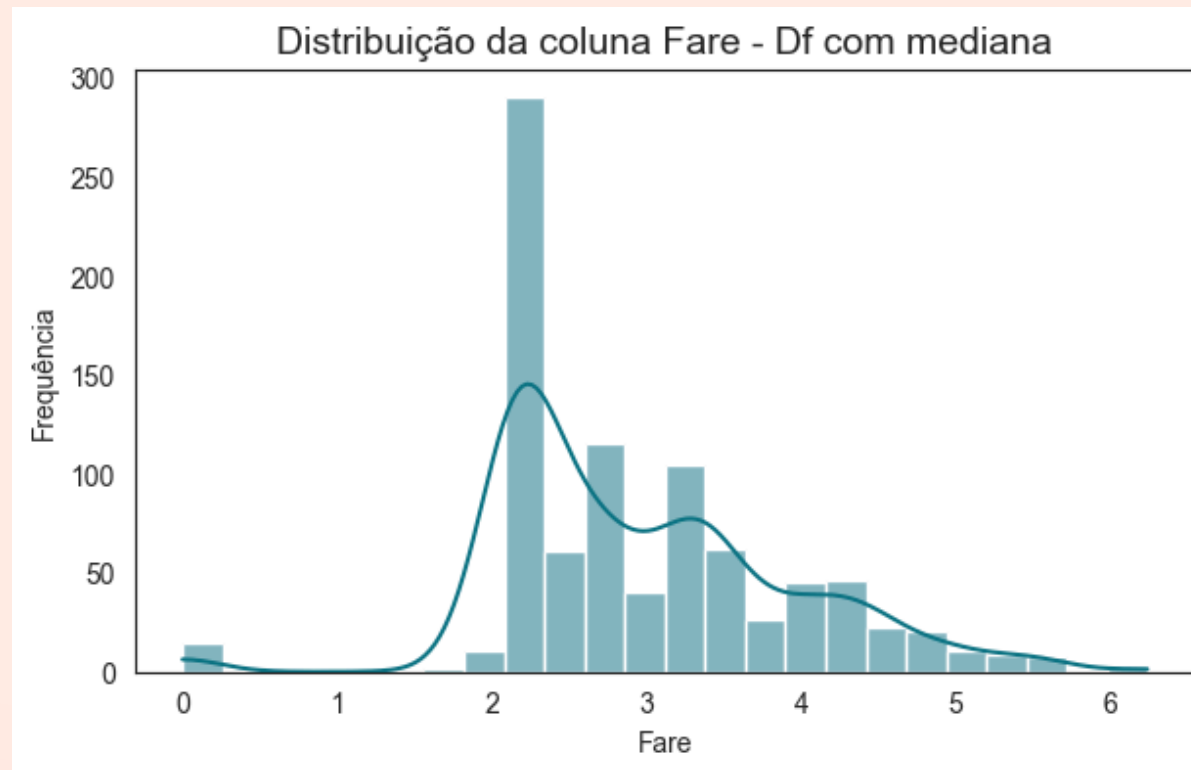
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QUESTÃO 2

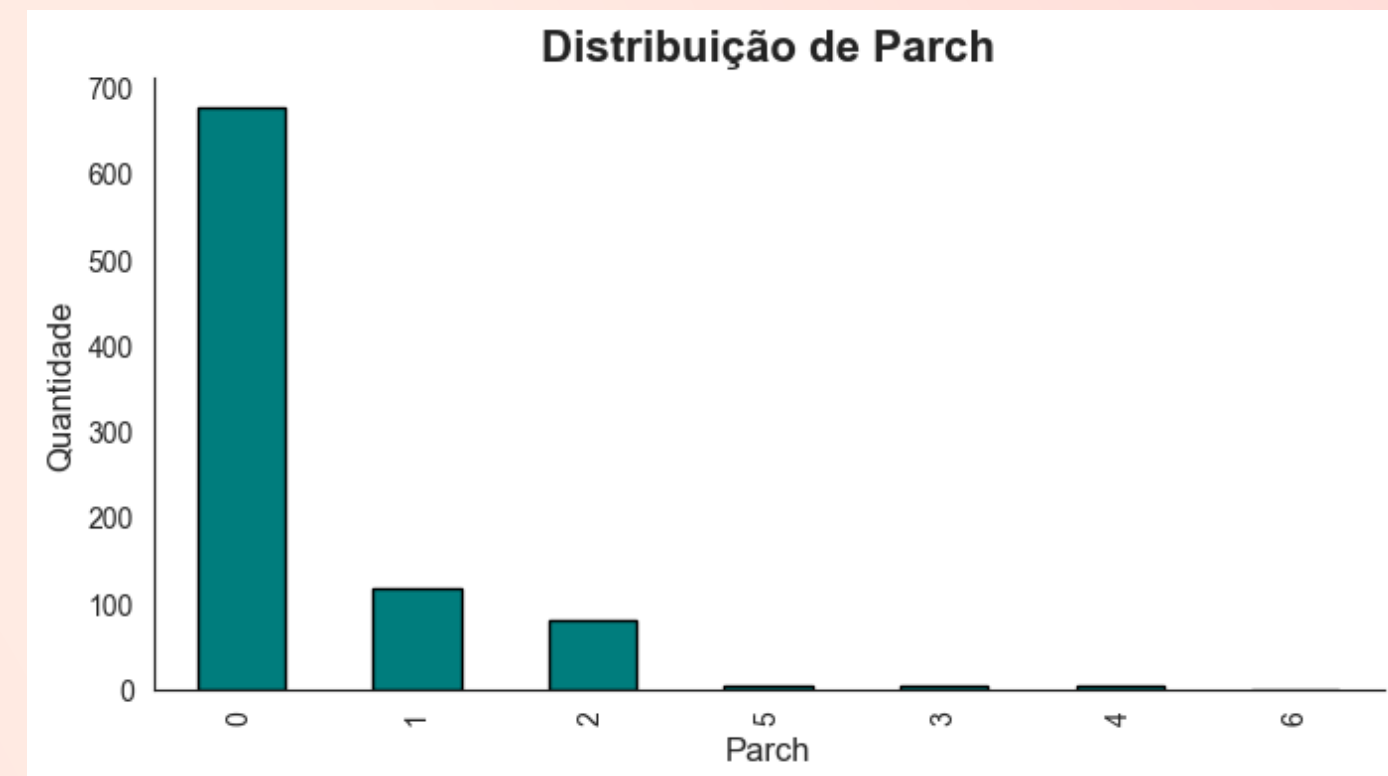
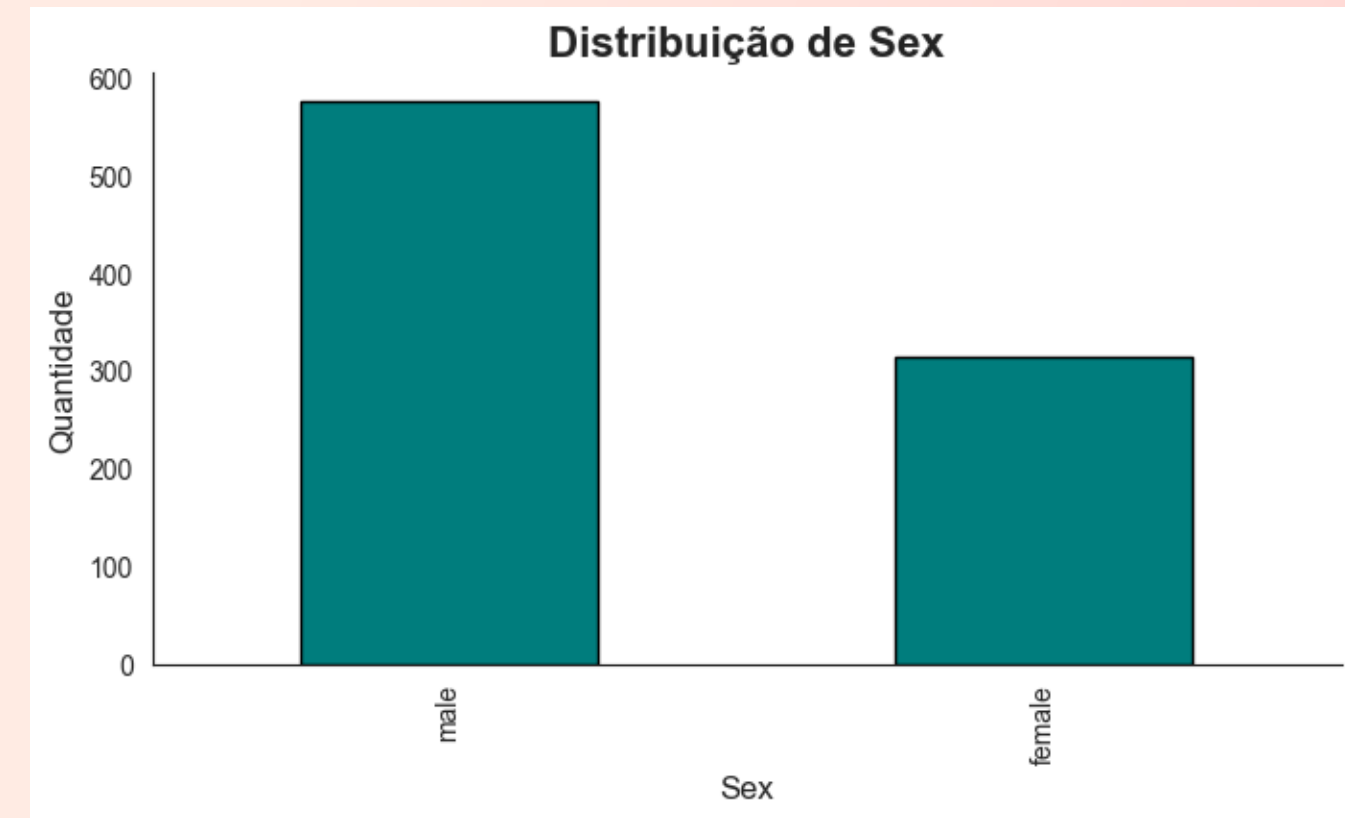
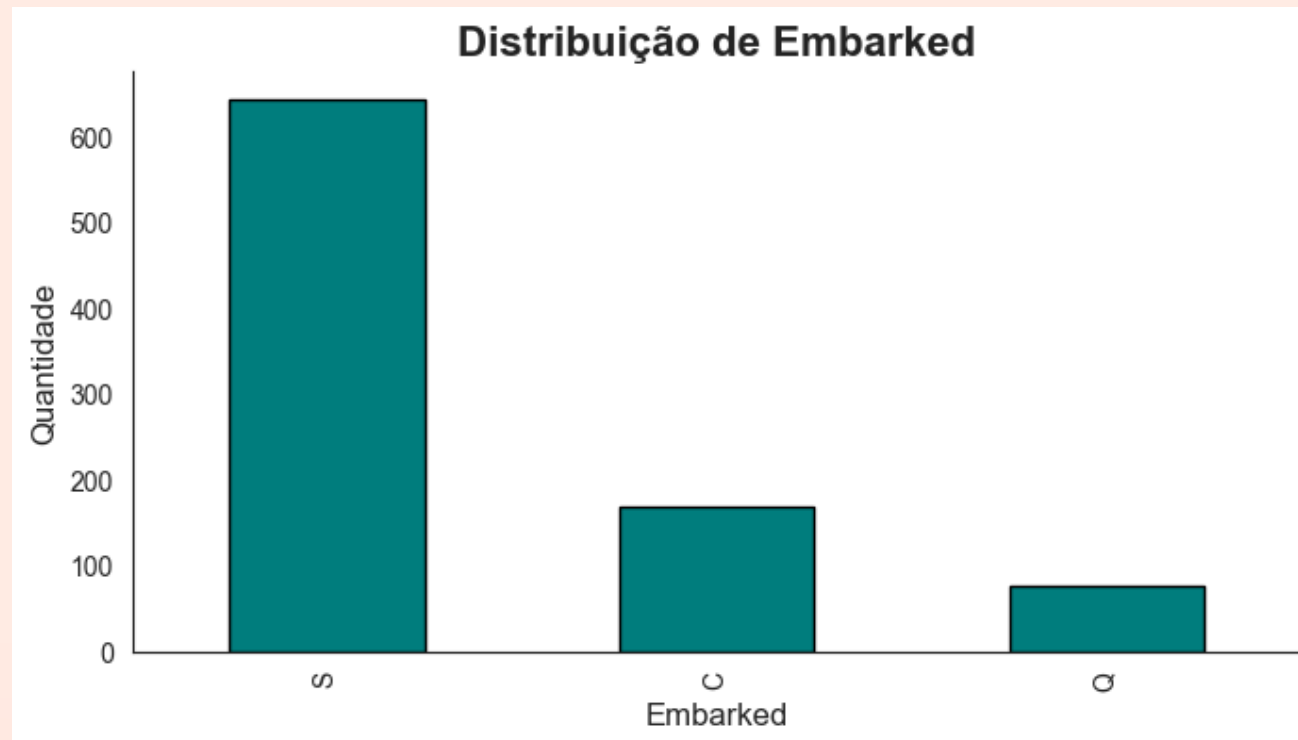
- Primeiramente, observou-se as colunas e quais valores elas continham.
- Foi escolhido retirar as colunas 'PassengerId', 'Name', 'Ticket', 'Cabin', tendo em vista que não apresentam nenhuma informação relevante para o treinamento.
- Observou-se que na coluna 'Age' apresentava muitos campos nulos, por isso, utilizou-se duas abordagens, uma retirando esses valores nulos e outra substituindo pela mediana. Além disso, foi balanceada pela coluna "Survived".



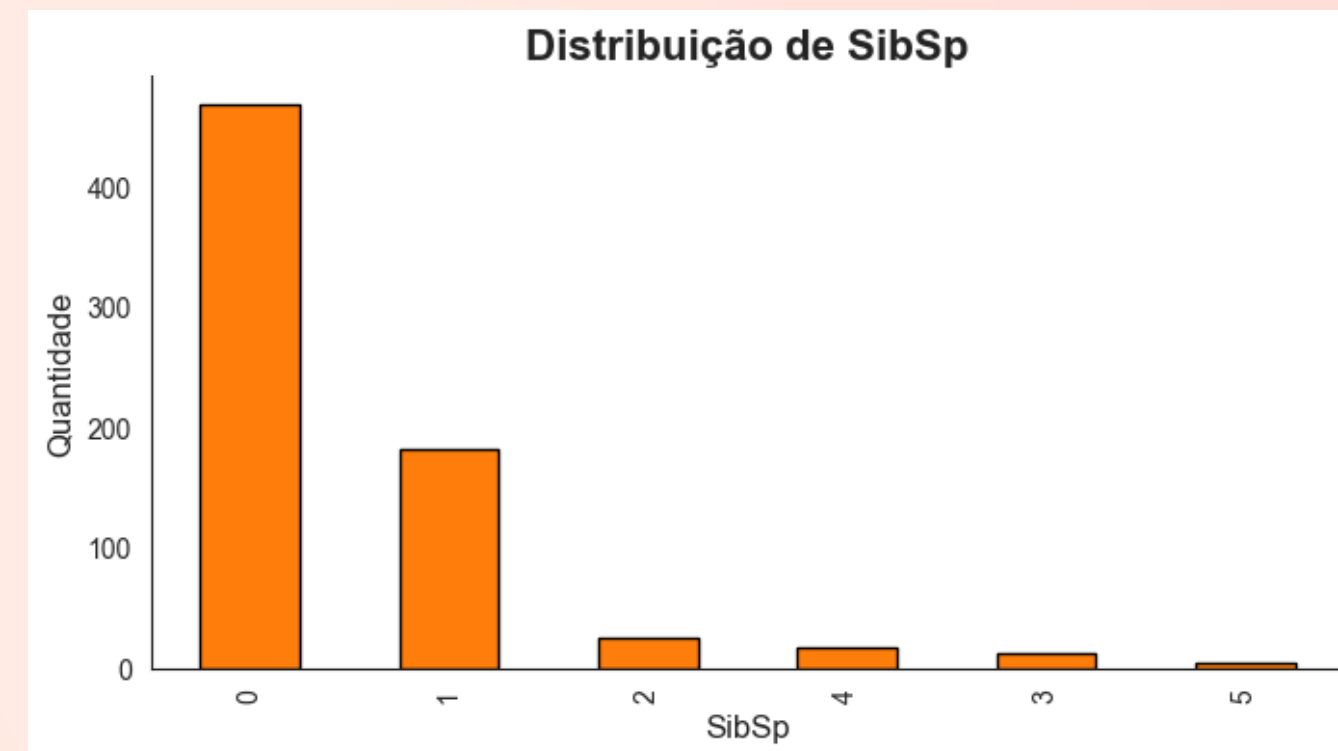
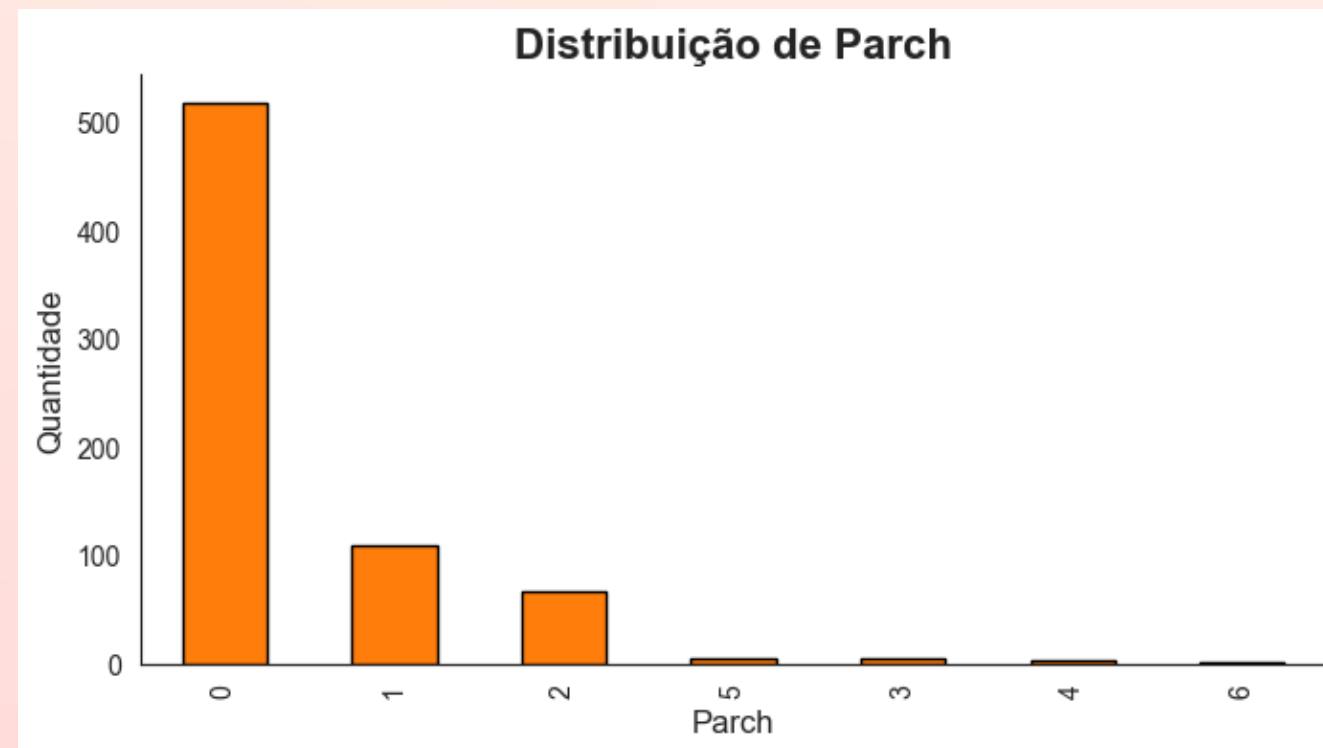
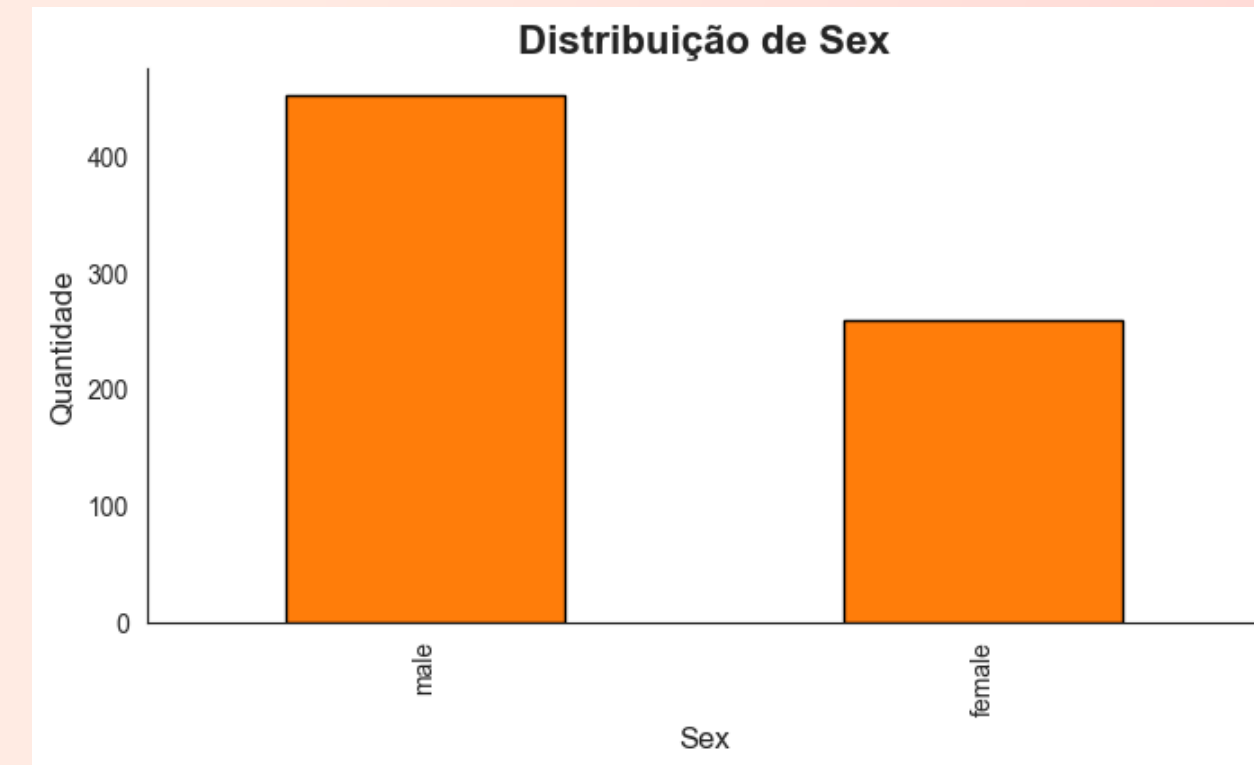
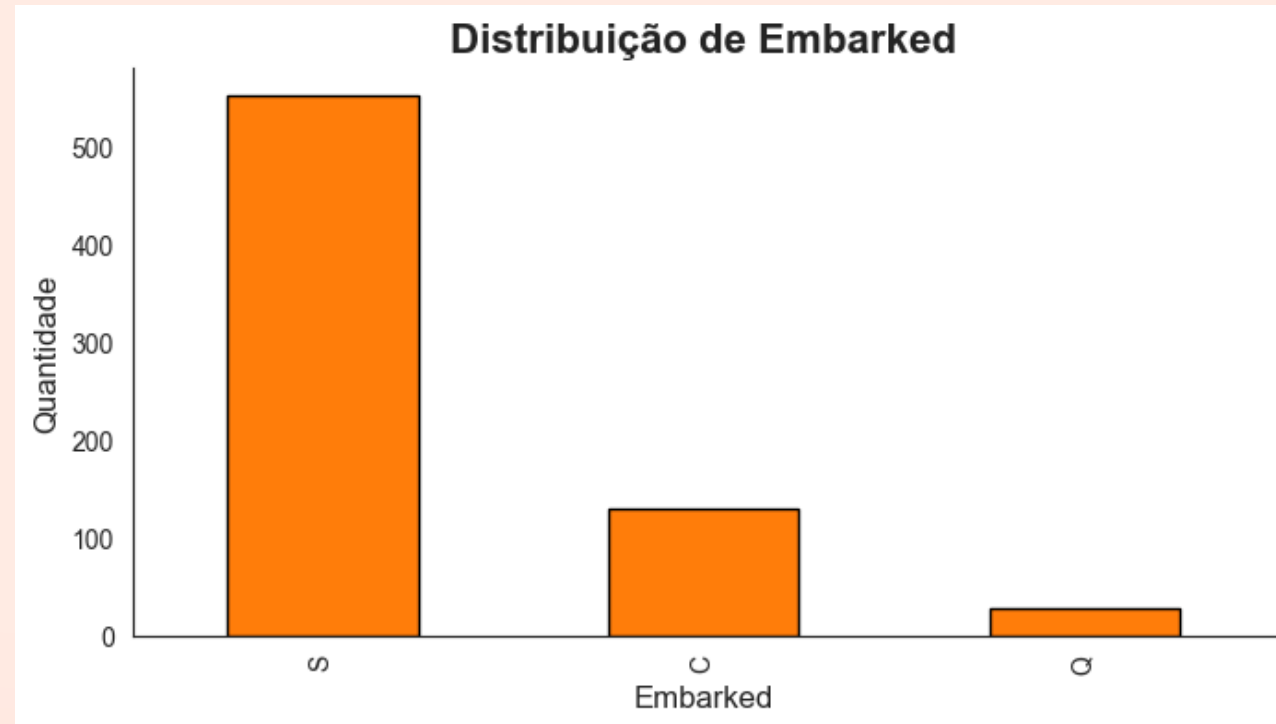
QUESTÃO 2



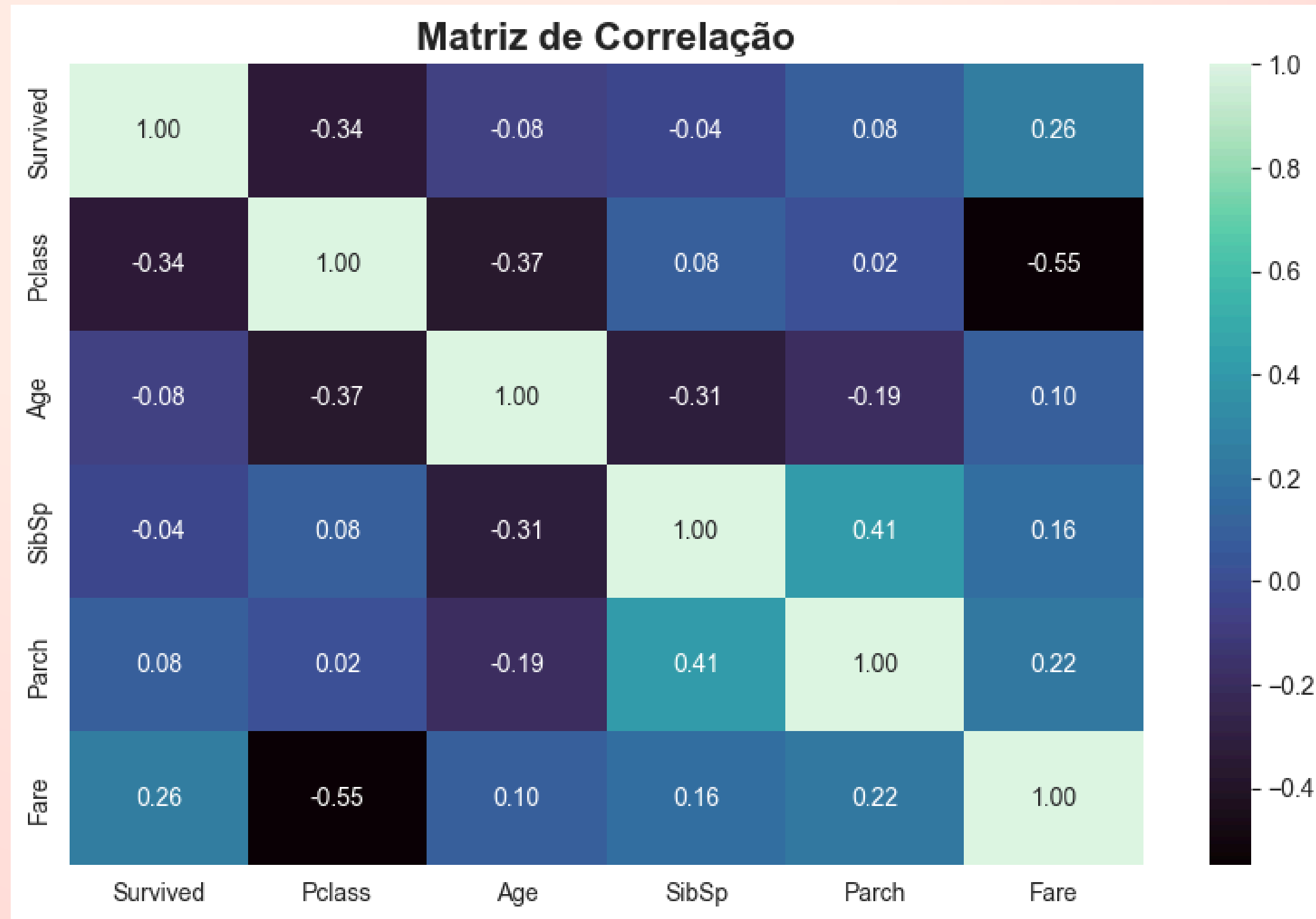
QUESTÃO 2



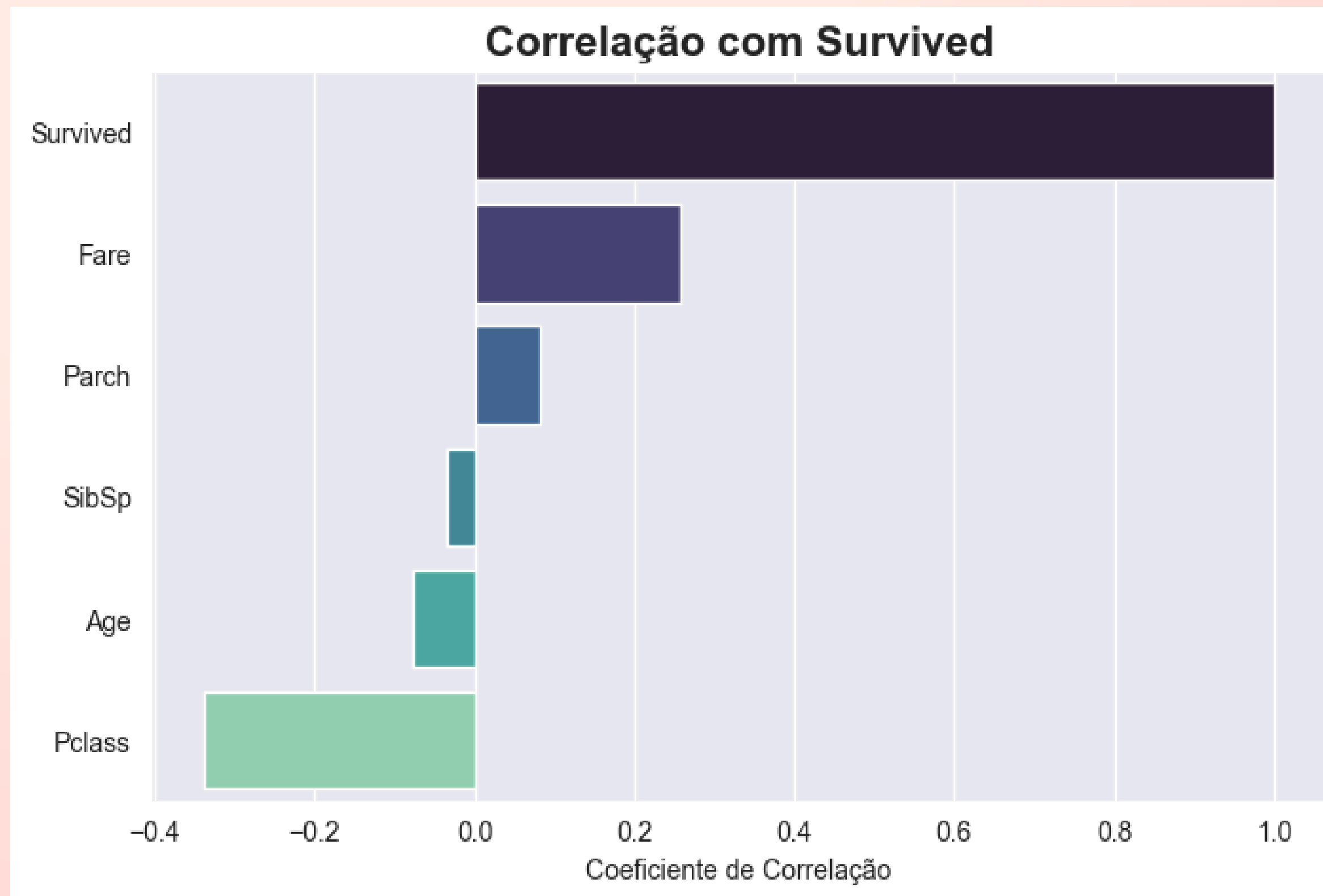
QUESTÃO 2



QUESTÃO 2

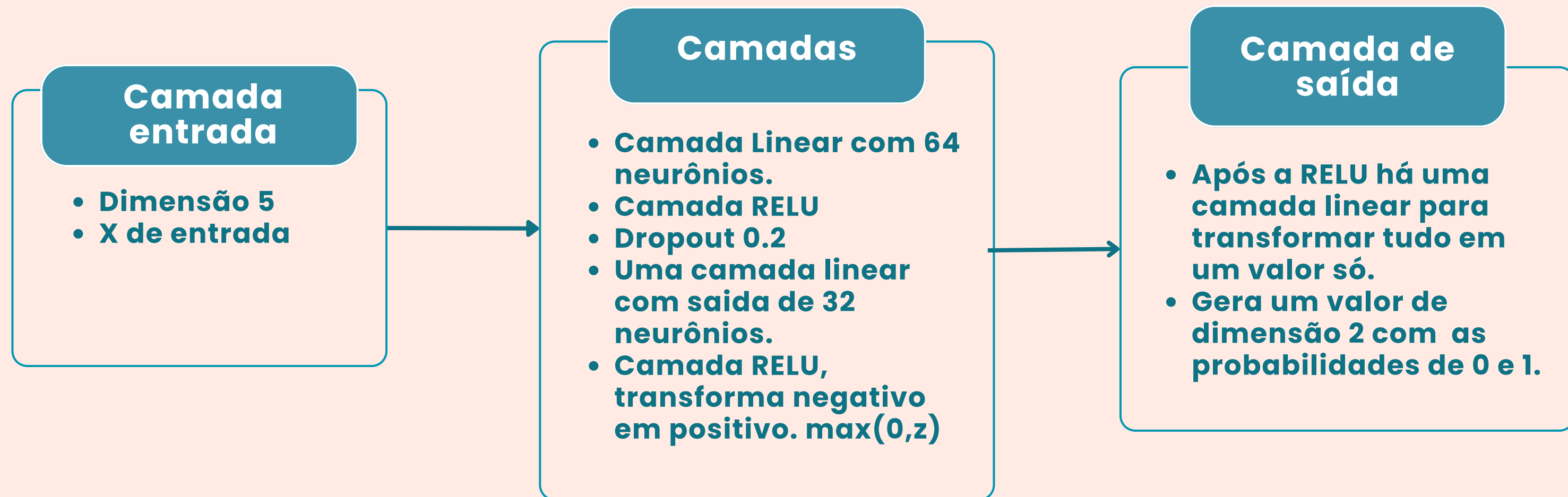


QUESTÃO 2



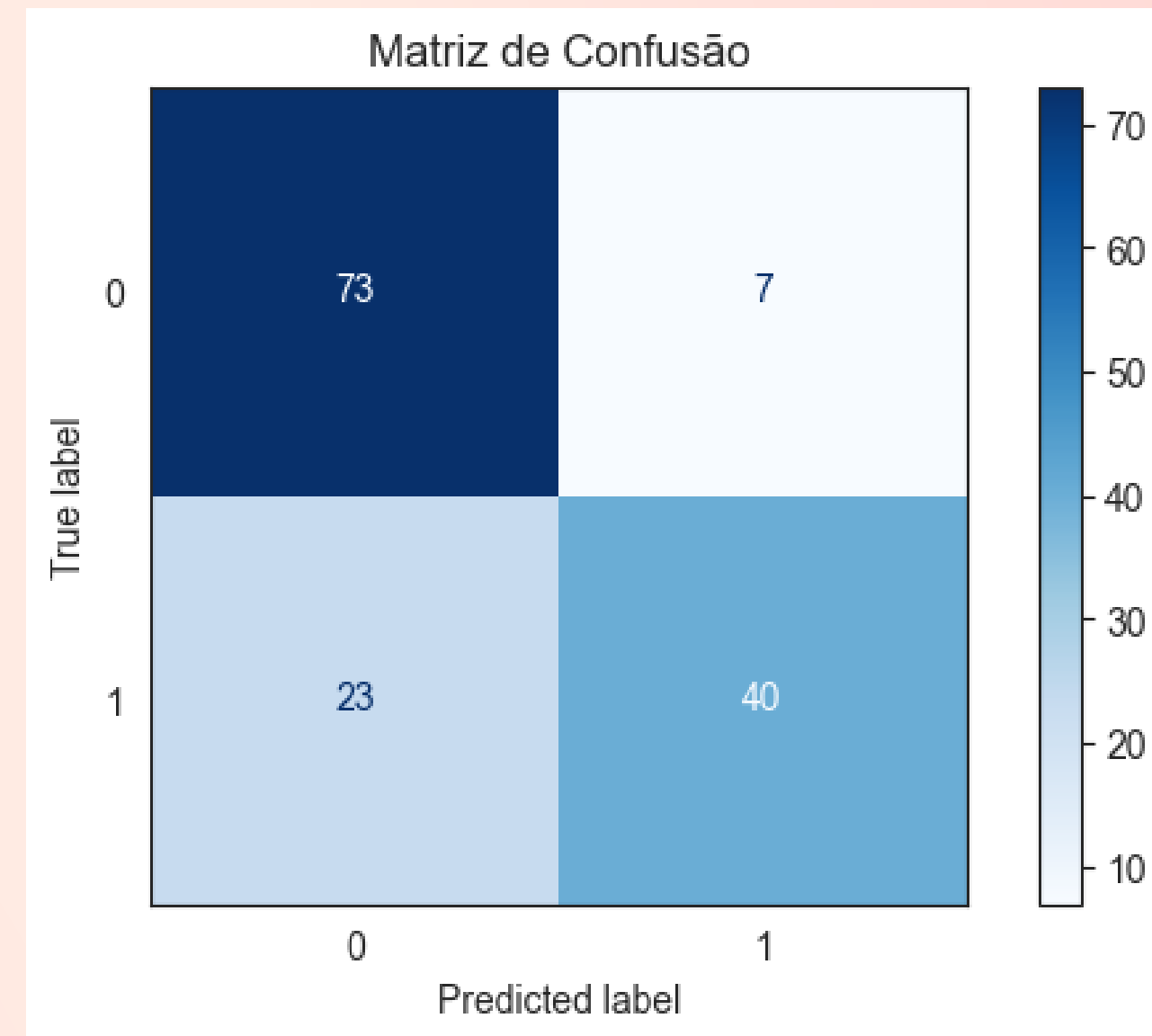
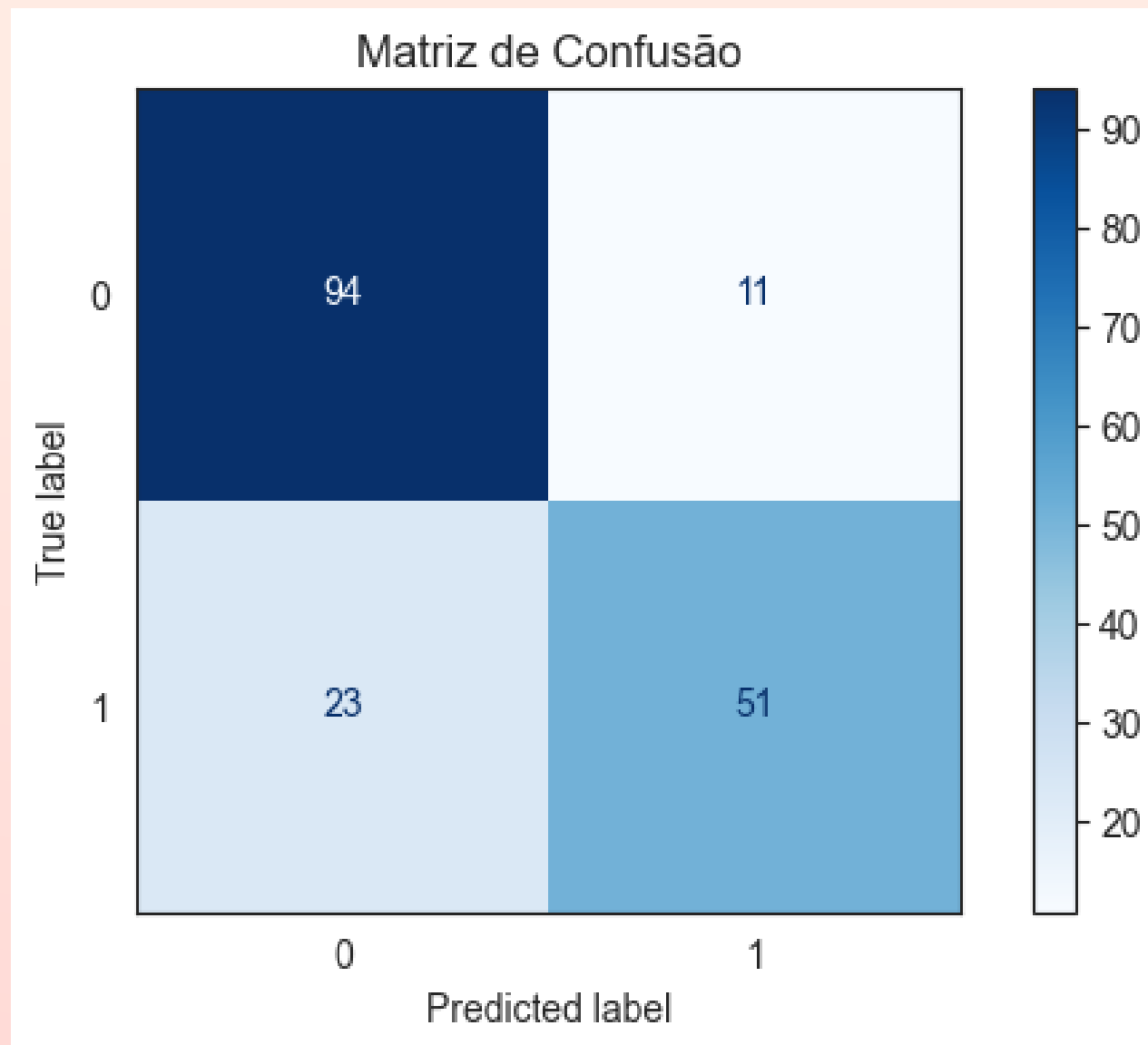
ARQUITETURA

Rede neural



QUESTÃO 2

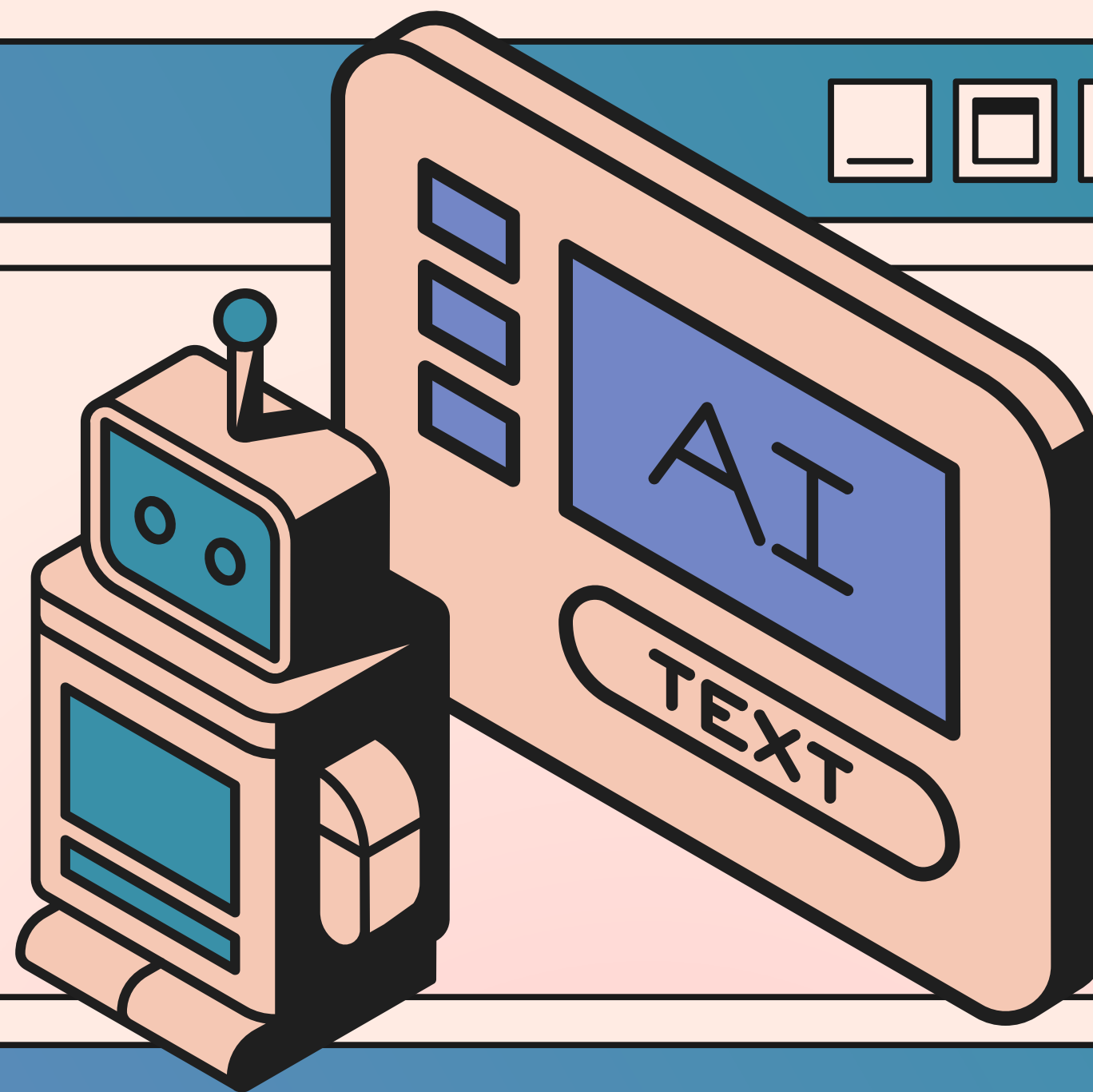
- Os dados da coluna "Age" e normalizada, e na coluna Fare foi utilizado log para obter trabalhar com os dados da melhor forma.
- Nas colunas com dados categóricos como "Sex" e "Embarked" foi utilizado o OneHotEncoder para utilizar essas variáveis no modelo.
- Obtém-se como resultado acurácia aproximada de 82% com mediana e 79% com drop para os casos, usando RN similar a questão 1, com dropout 0.2.





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QUESTÃO 3



GISELE MENEZES , MARIANA MARTINS E THIAGO RODRIGUES

REDE NEURAL CONVOLUCIONAL (CNN)

QUESTÃO 3



CONJUNTO DE DADOS CIFAR-10

- Possui **60.000** imagens coloridas **32x32** divididas em **10** classes.
 - 50.000 imagens para treinamento e 10.000 imagens para teste.
- **load_cifar10_numpy** - carrega os arquivos originais do CIFAR-10 (.bins) e organiza em arrays do NumPy
 - Reshape para $(N, 3, 32, 32) \rightarrow (N, C, H, W)$

```
print("Classes:", classes)
print(f"\nX_train: {X_train.shape}, {X_train.dtype}")
print(f"y_train: {y_train.shape}, {y_train.dtype}")
print(f"X_test : {X_test.shape}, {X_test.dtype}")
print(f"y_test : {y_test.shape}, {y_test.dtype}")
```

[65]

```
... Classes: ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']
```

```
X_train: (50000, 3, 32, 32), uint8
y_train: (50000,), int64
X_test : (10000, 3, 32, 32), uint8
y_test : (10000,), int64
```

airplane



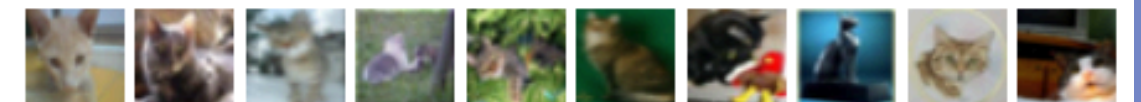
automobile



bird



cat



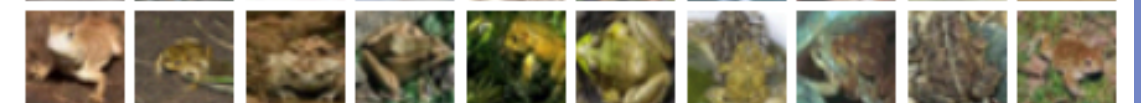
deer



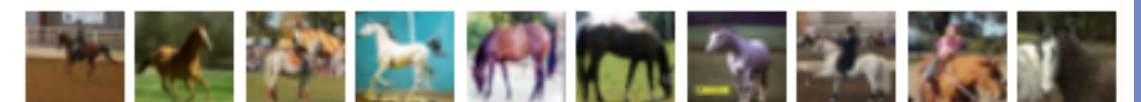
dog



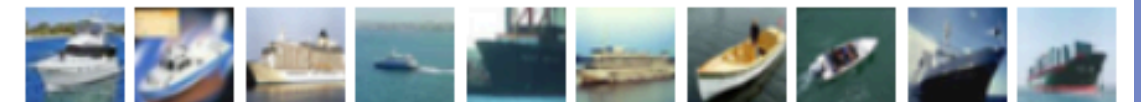
frog



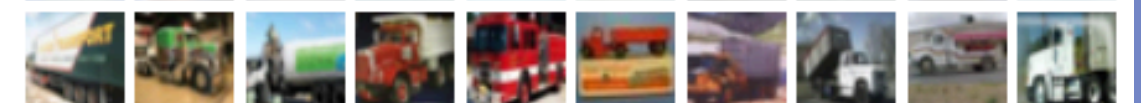
horse



ship



truck



CONJUNTO DE VALIDAÇÃO

- **stratified_train_val_split(...)**
 - Cada classe teve $\approx 10\%$ das amostras movidas para validação;
 - Evita que uma classe fique sub ou super-representada;

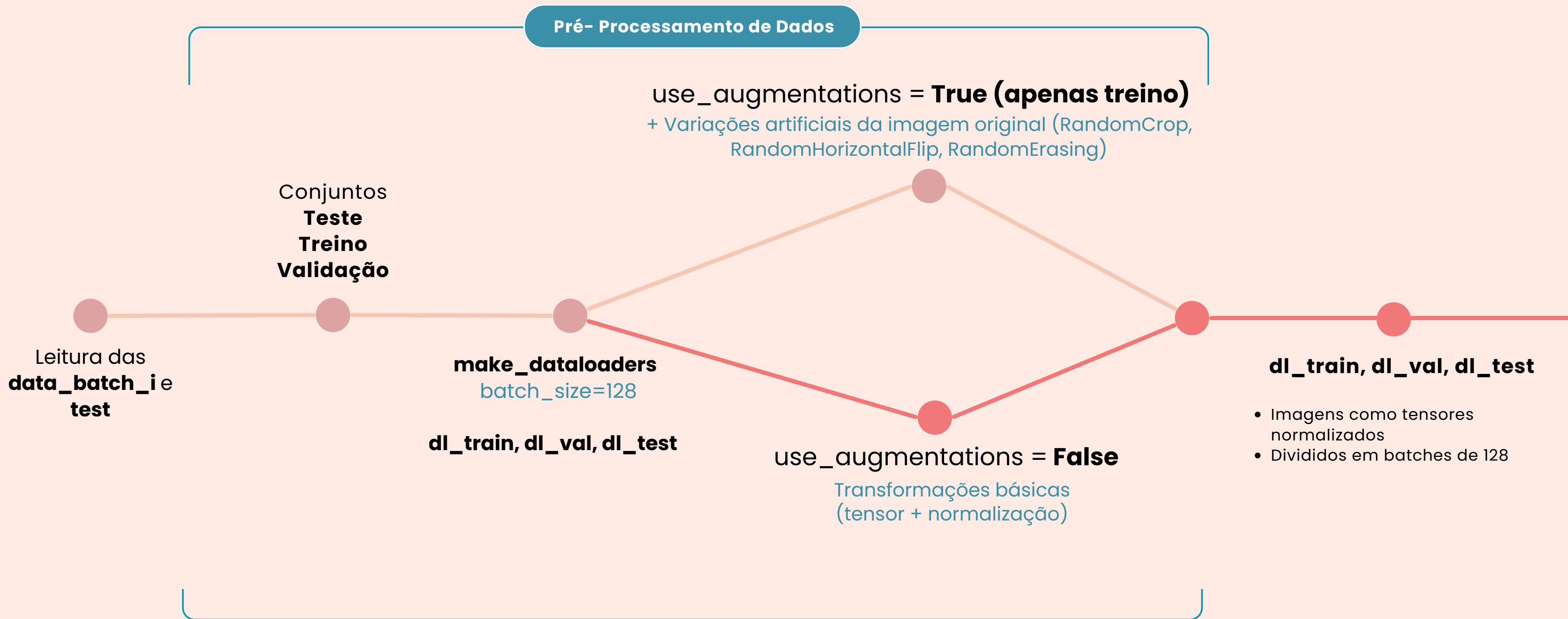
Teste: 10.000 imagens

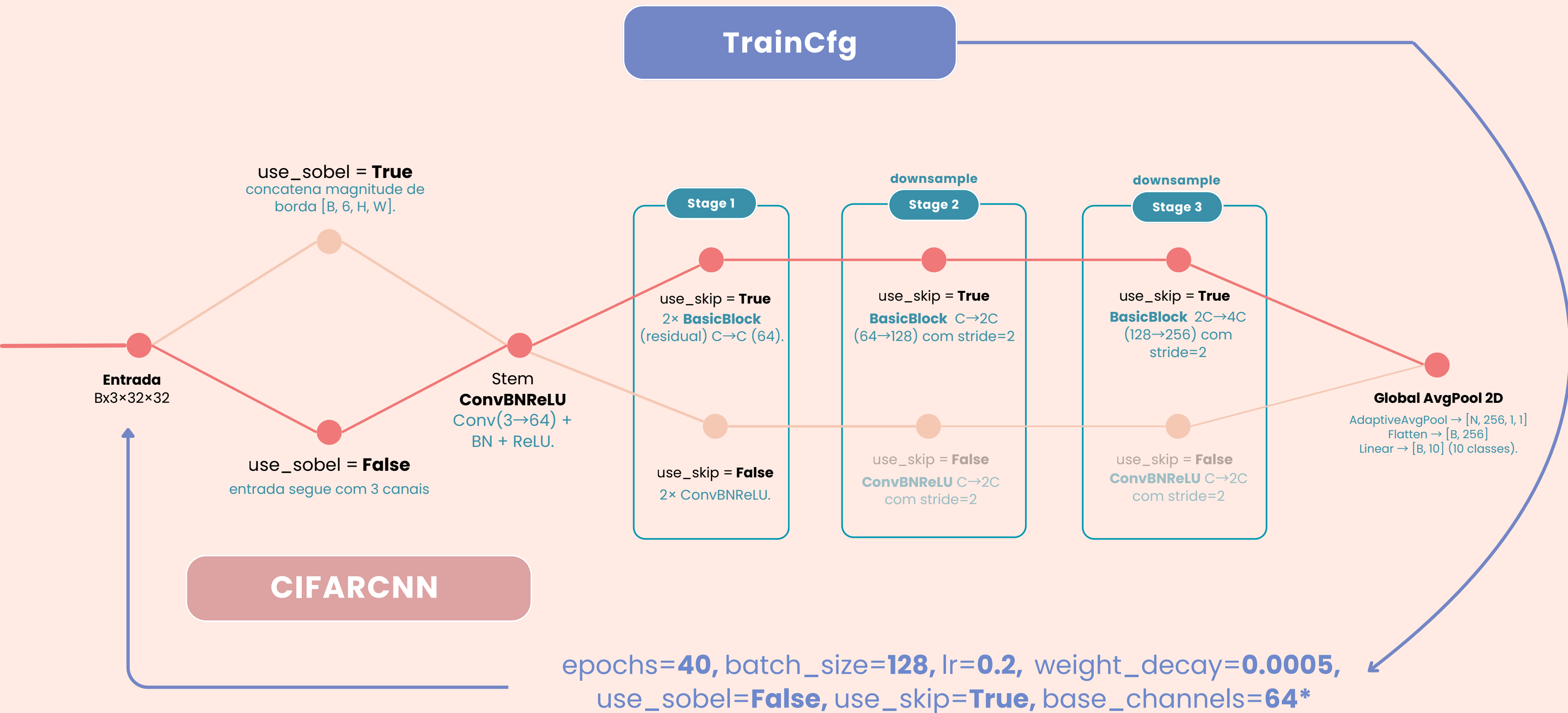
Treino: **45.000 imagens**

Validação: **5.000 imagens**

PIPELINE

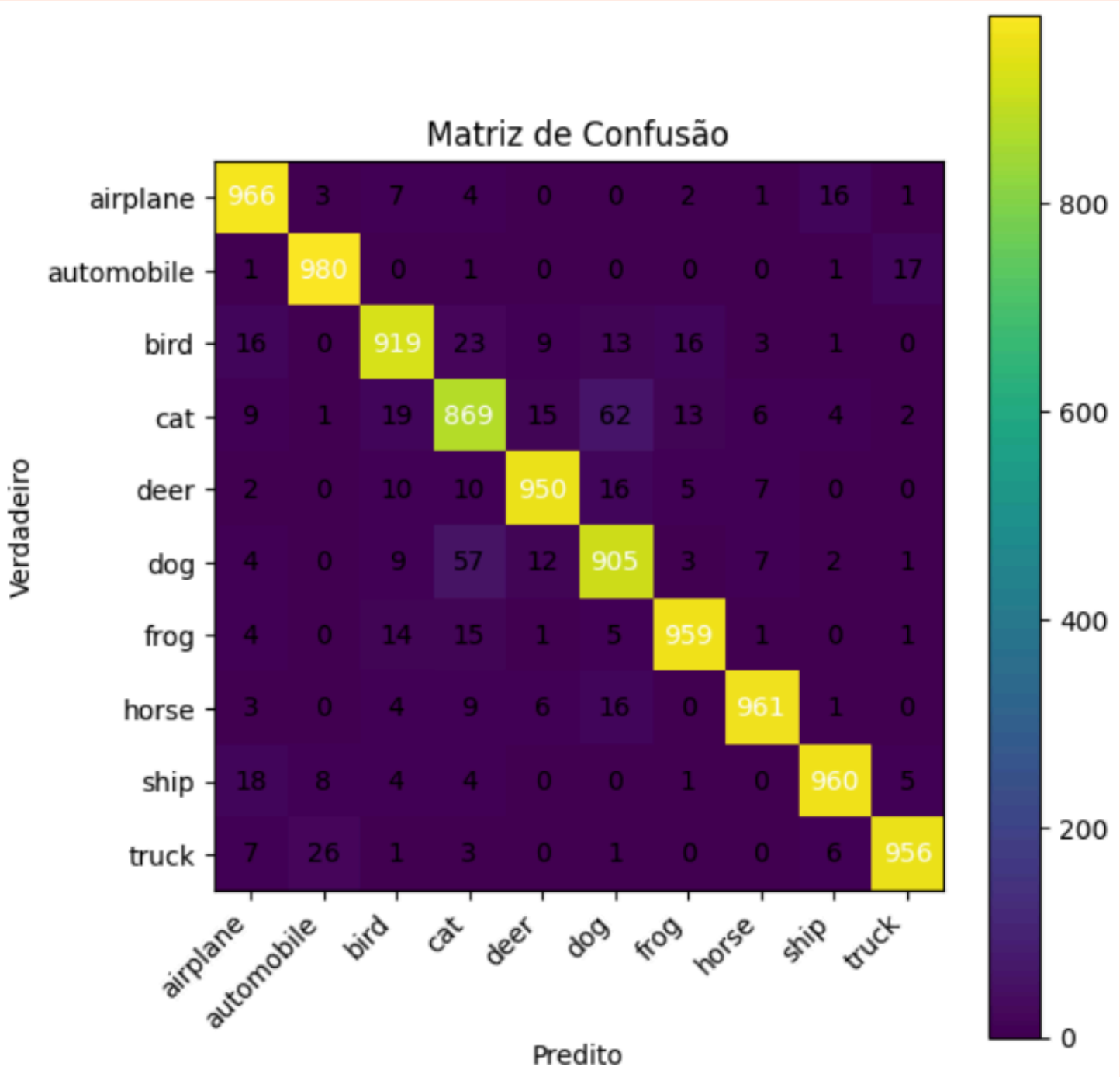
& FLUXO





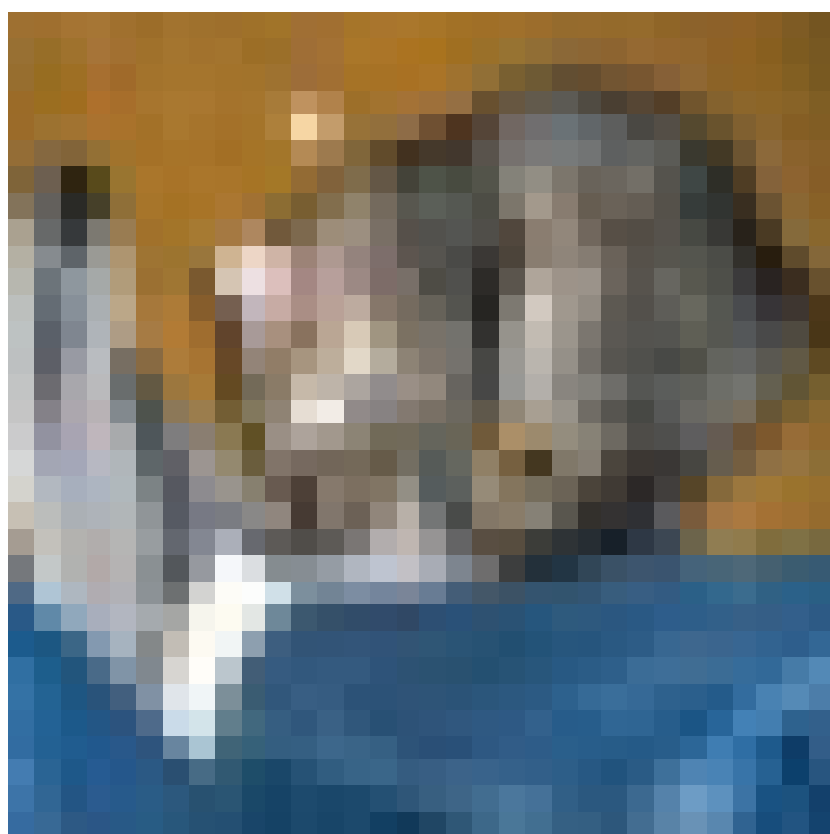
RESULTADOS

Acurácia global: 94.25%



airplane	96.60%
automobile	98.00%
bird	91.90%
cat	86.90%
deer	95.00%
dog	90.50%
frog	95.90%
horse	96.10%
ship	96.00%
truck	95.60%

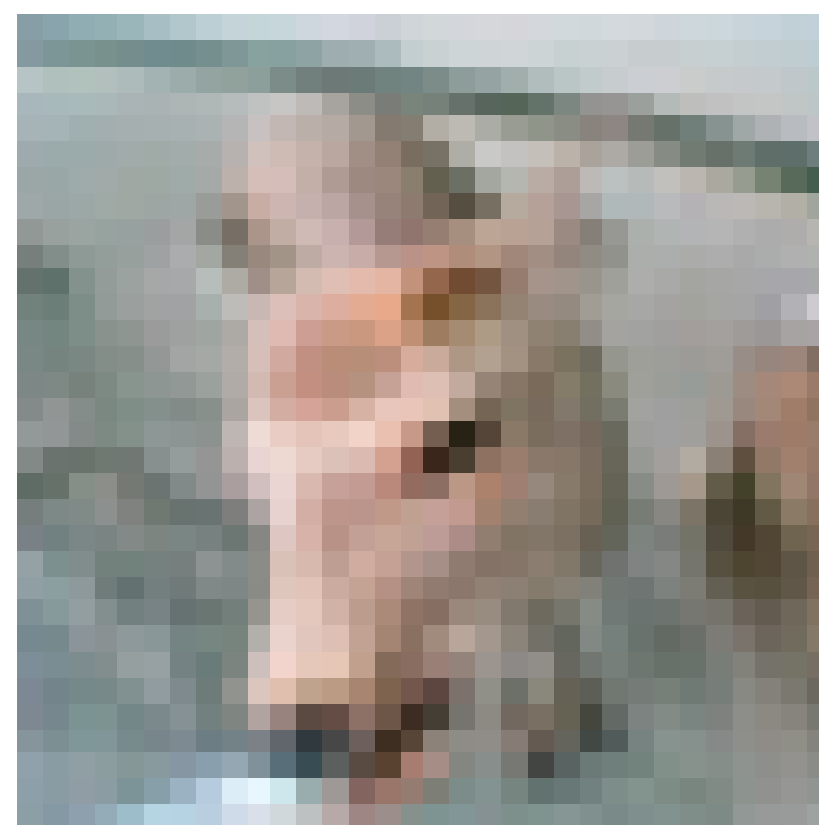
Acerto #1
Pred: cat (1.00)
True: cat



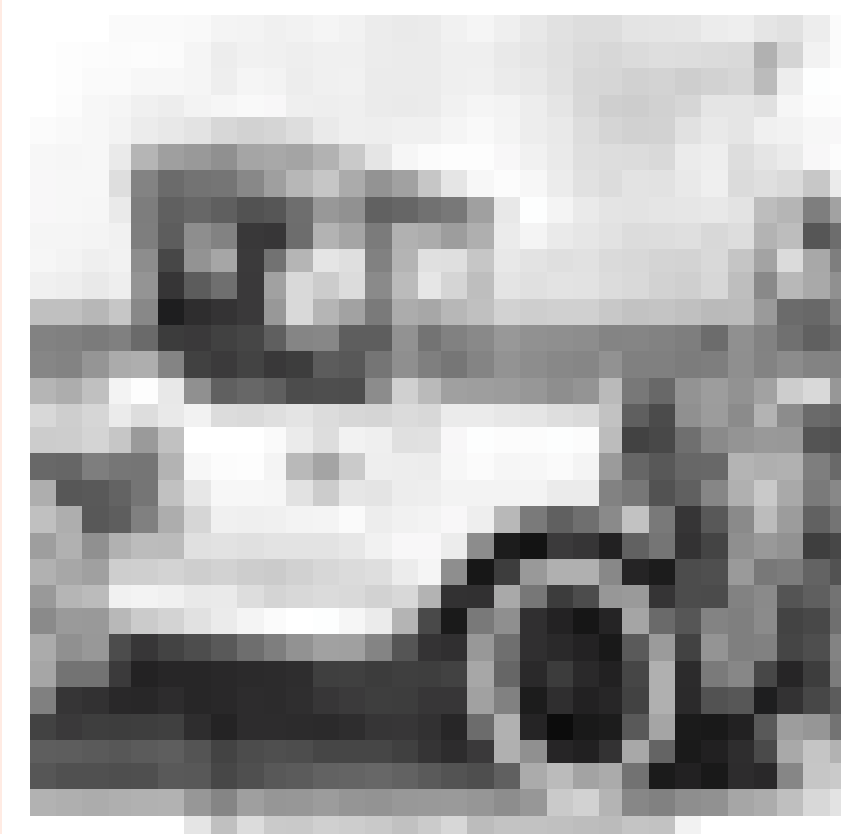
Acerto #4
Pred: airplane (0.97)
True: airplane



Erro #4
Pred: dog (0.81)
True: deer

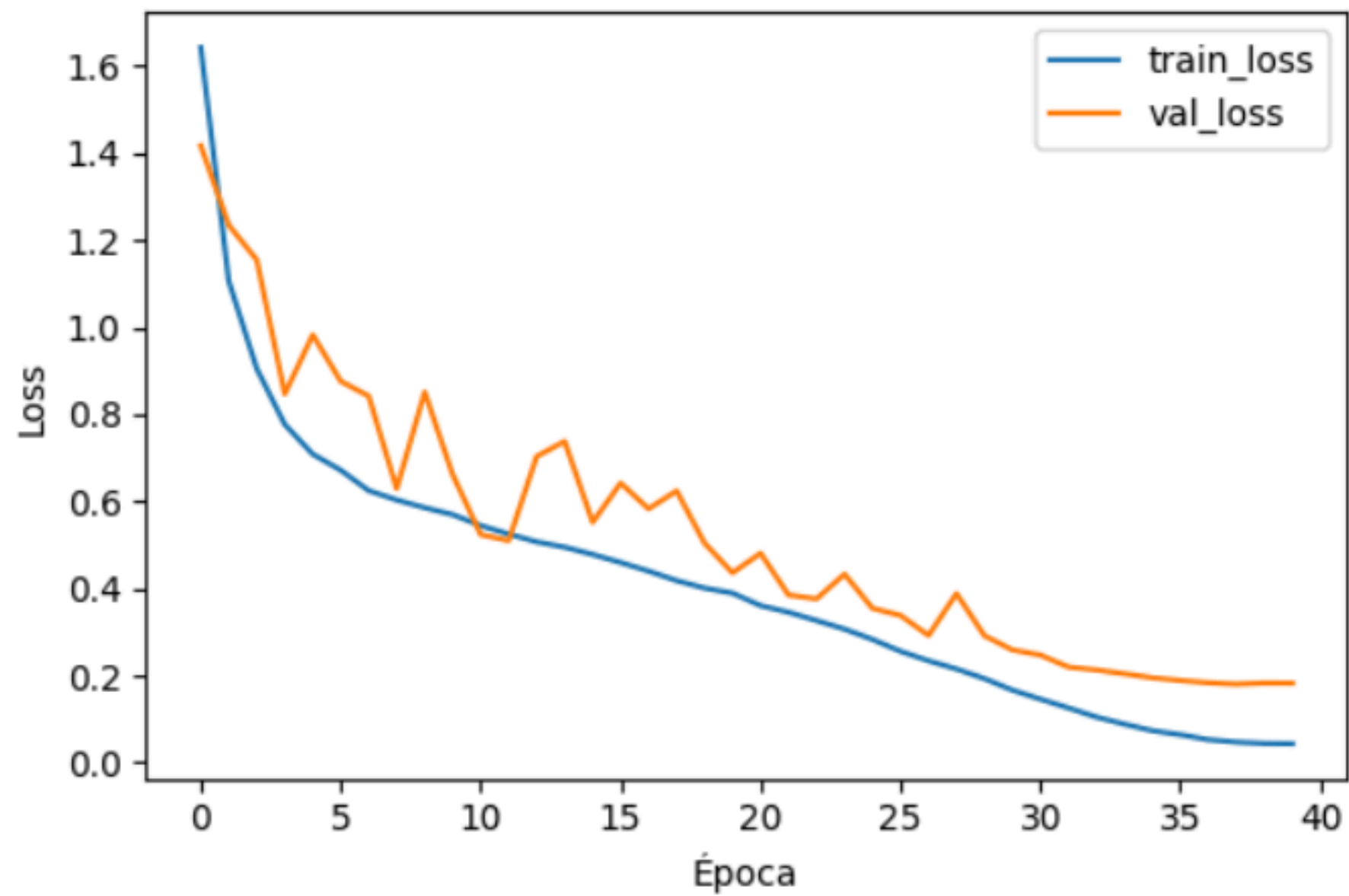


Erro #2
Pred: truck (0.80)
True: automobile

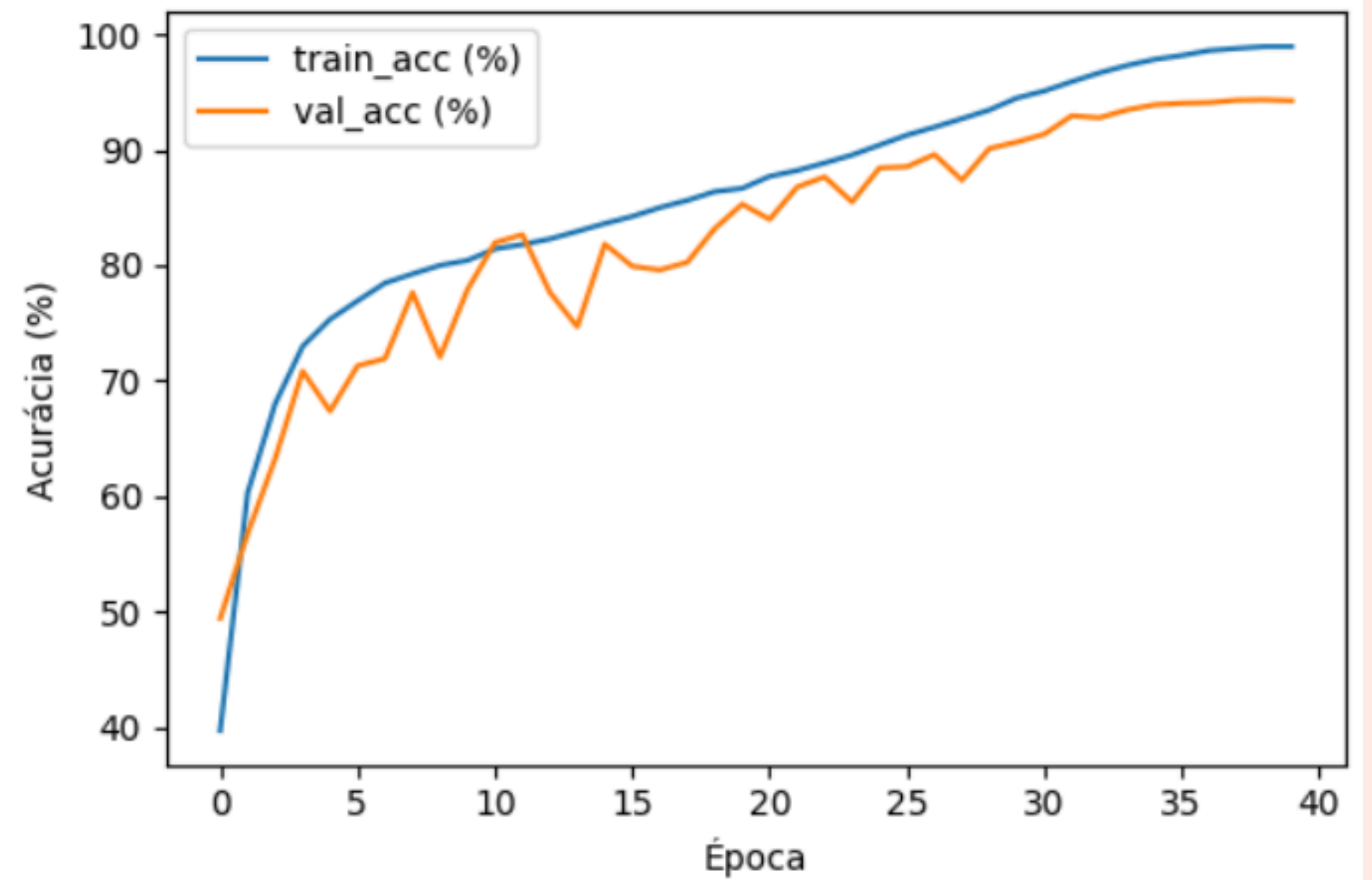


RESULTADOS

Histórico de Loss



Histórico de Acurácia

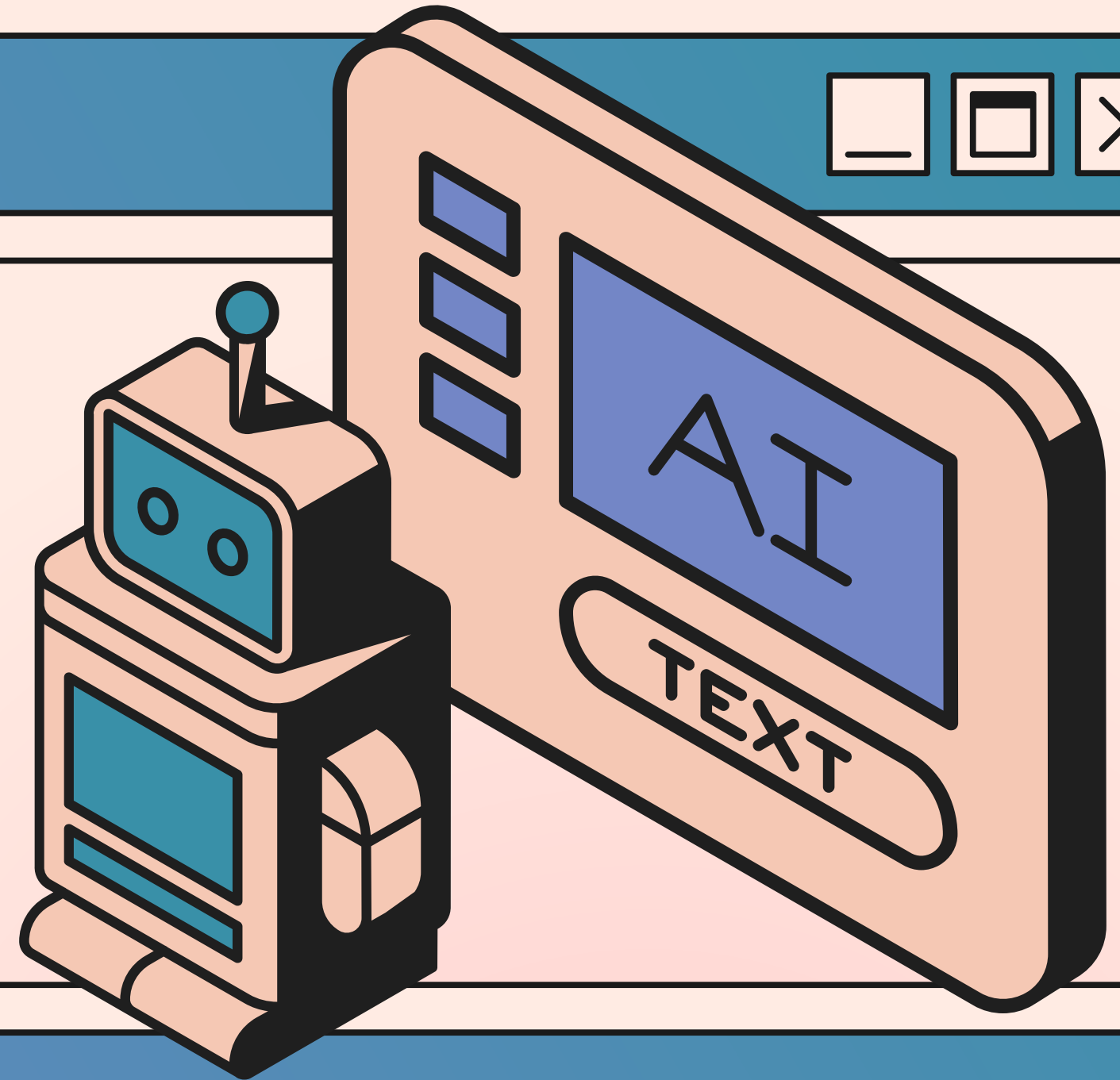


Treino	Épocas	Batch	LR	Sobel	Skip	Acurácia (Teste)	Obs.
11	40	128	0.2	✗	✓	94.25%	
12	40	128	0.2	✓	✓	93.78%	—
5	40	128	0.2	✗	✗	93.13%	—
4	30	128	0.2	✗	✗	92.46%	—
8	30	256	0.2	✗	✗	92.44%	—
9	40	256	0.2	✗	✗	92.38%	—
10	40	256	0.2	✗	✗	91.92%	Config resumida
Com Aug	30	128	0.2	✗	✗	91.79%	Data Augmentation
3	30	128	0.1	✗	✗	91.72%	—
2	20	128	0.1	✗	✗	91.27%	—
7	20	256	0.1	✗	✗	90.99%	—
Extra	20	264	0.1	✗	✗	90.82%	Batch diferente
6	20	128	0.1	✓	✗	90.20%	Sobel ativo
1	2	128	0.1	✗	✗	73.58%	CPU



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DEEP LEARNING - 2025.1

QUESTÃO 4

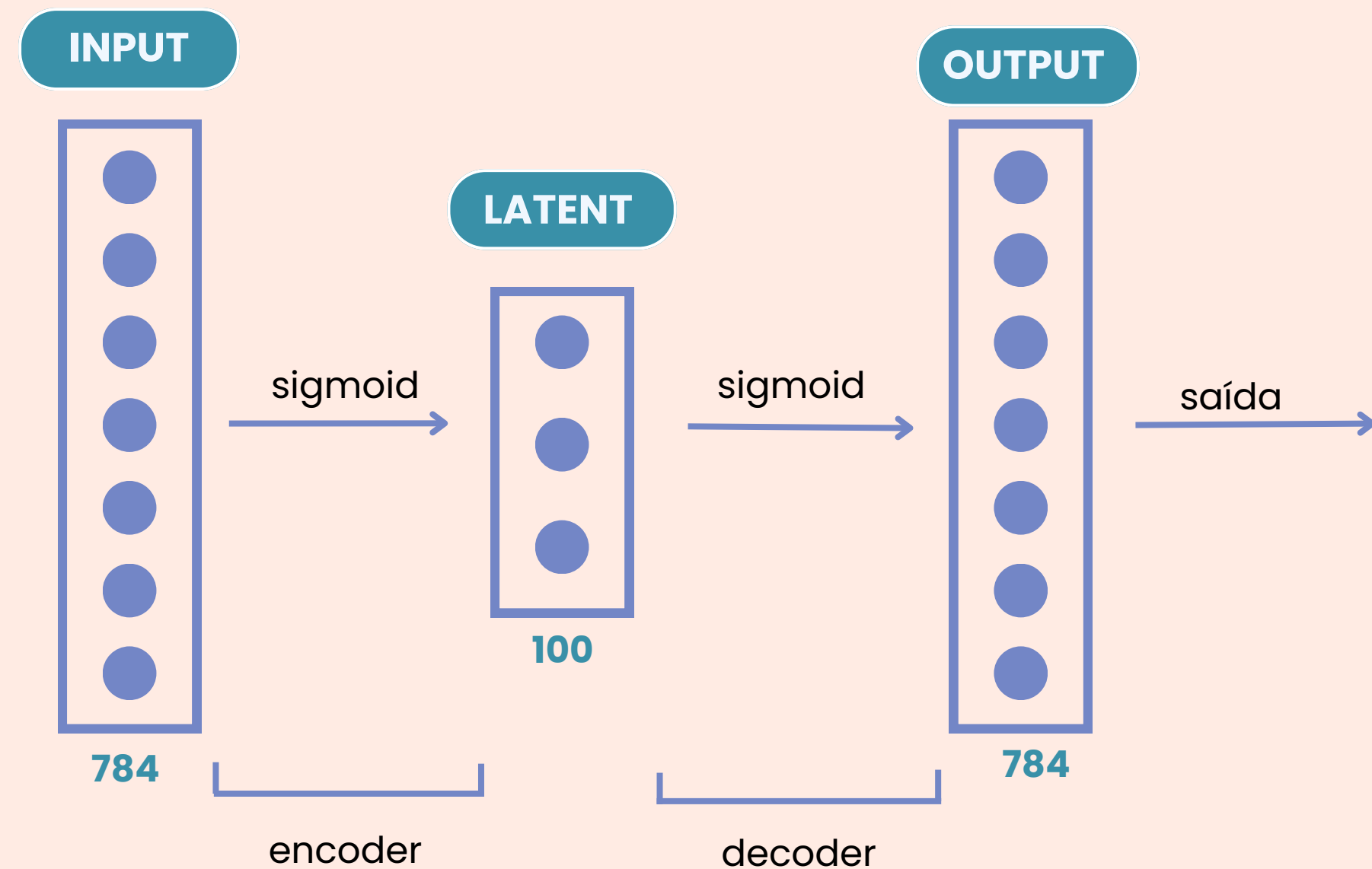


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QUESTÃO 4 - A

- **Camada de entrada:**
 - Dimensão: $\text{input_dim} = 28 \times 28 = 784$
 - Recebe os pixels da imagem do Fashion-MNIST
- **Camada latente (encoder):**
 - Dimensão: $\text{latent_dim} = 100$
 - Aplica sigmoid sobre a comb. linear da entrada.
- **Camada de saída (decoder):**
 - Dimensão: $\text{input_dim} = 784$
 - Aplica sigmoid sobre a comb. linear da camada latente para reconstruir a imagem.

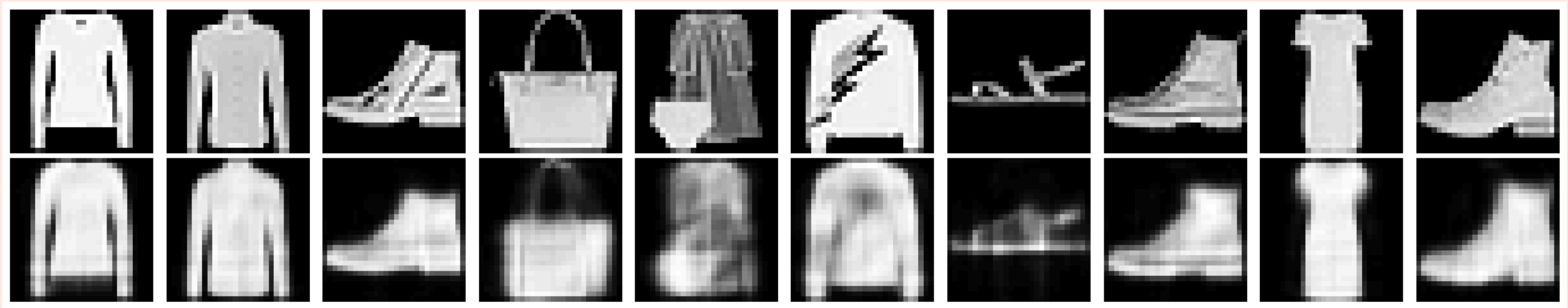
AUTOENCODER



OBJETIVO: RECONSTRUIR A IMAGEM ORIGINAL



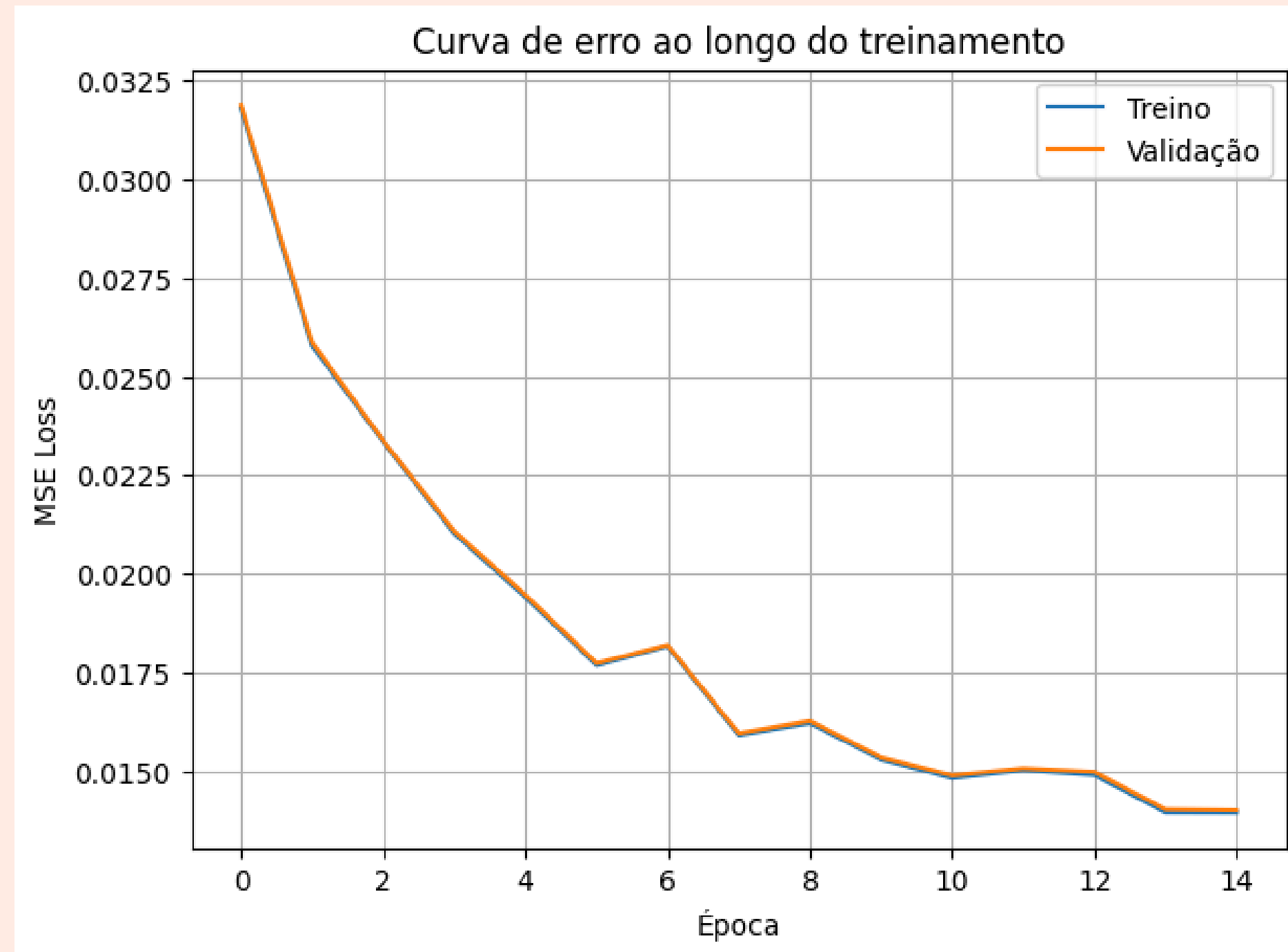
IMAGEM ORIGINAL X IMAGEM RECONSTRUÍDA



QUESTÃO 4-A

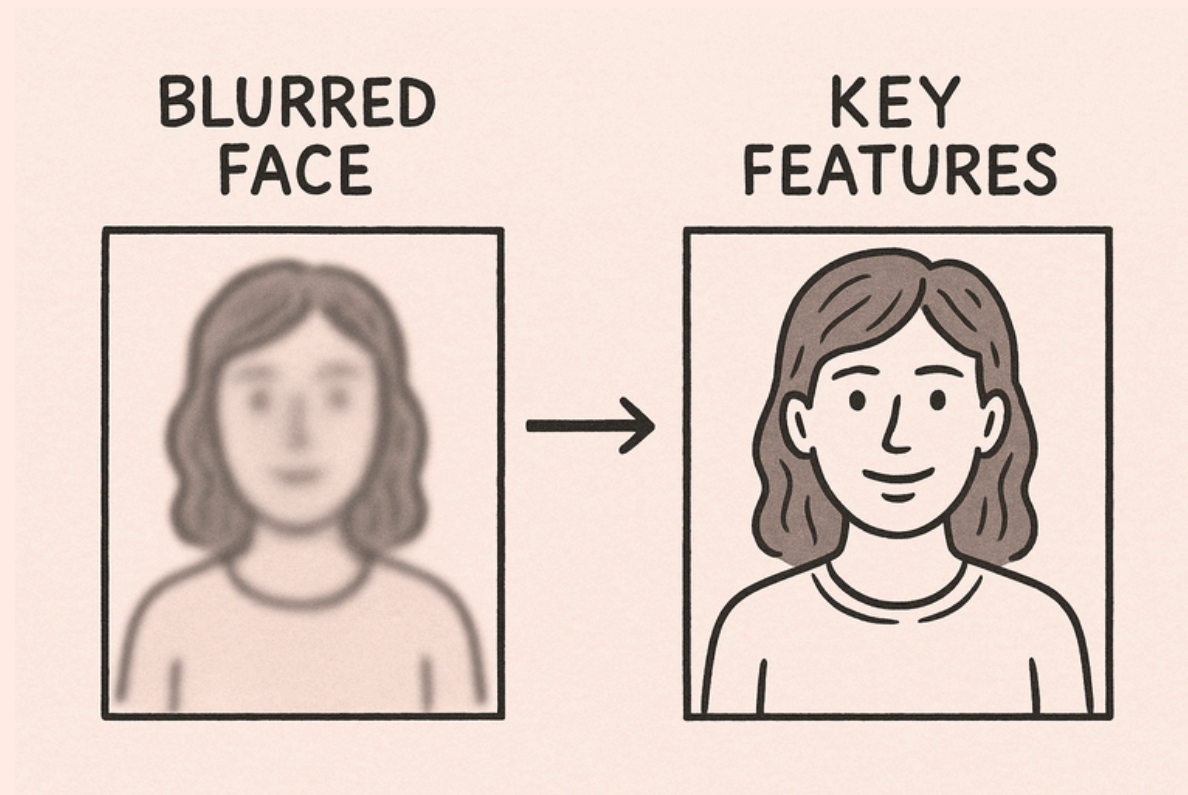
CURVA DE ERRO

QUESTÃO 4 - A



DENOISING AUTOENCODER

Exemplo:



1) def add_noise(X, noise_factor= 0.3):

```
X_noisy = X + noise_factor * np.random.randn(*X.shape)
X_noisy = np.clip(X_noisy, 0., 1.)
return X_noisy
```

2) Forward e Backward

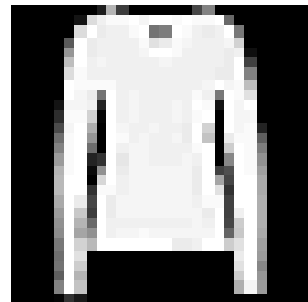
```
z, x_hat, z_lin, x_hat_lin = forward(x_noisy_batch,
W_enc, b_enc, W_dec, b_dec)
W_enc, b_enc, W_dec, b_dec = backward(x_batch, z,
x_hat, z_lin, x_hat_lin, ...)
```

OBJETIVO: RECONSTRUIR A VERSÃO LIMPA DA IMAGEM

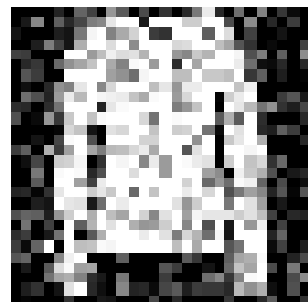


IMAGEM ORIGINAL x COM RUÍDO x RECONSTRUÍDA

Original



Com ruído



Reconstruída

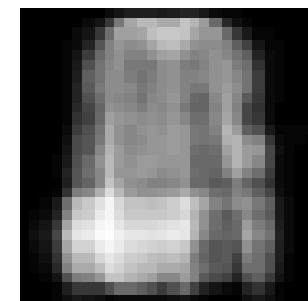
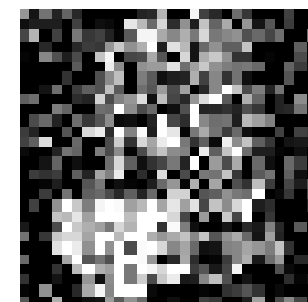
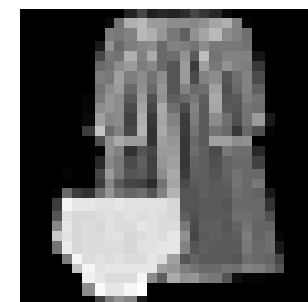
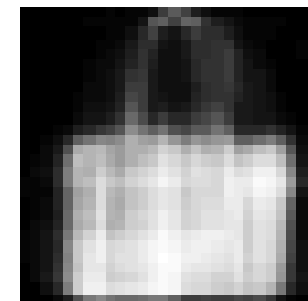
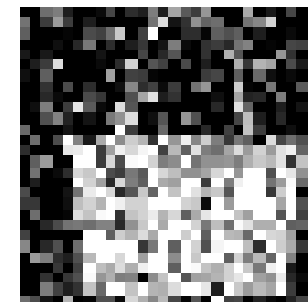
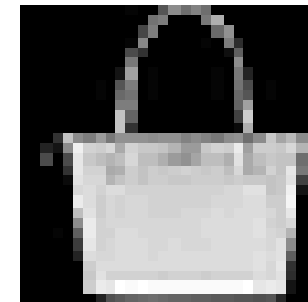
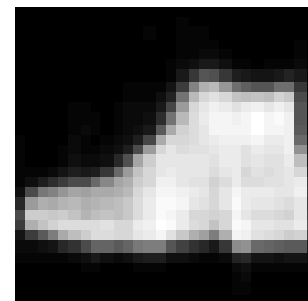
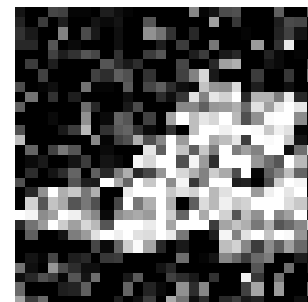
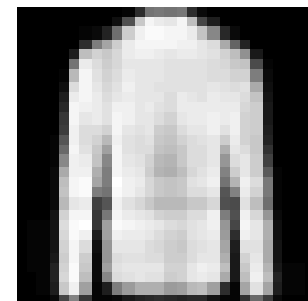
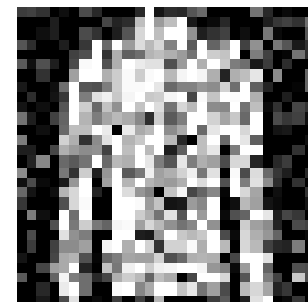
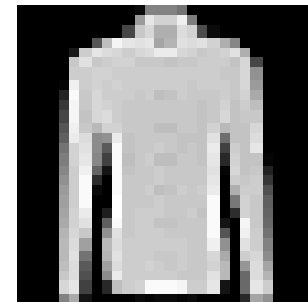
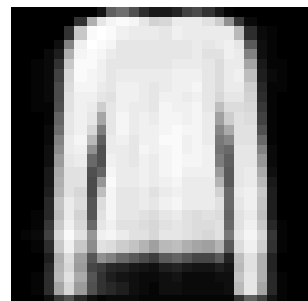
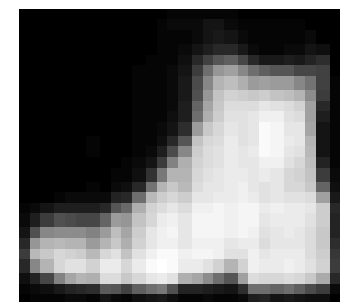
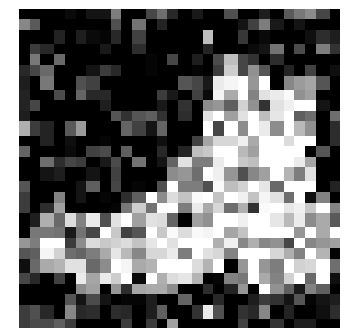
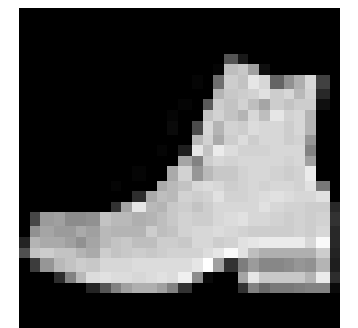
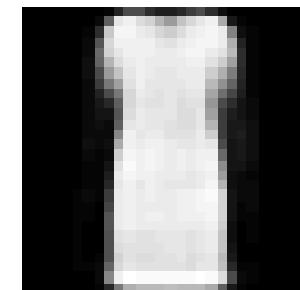
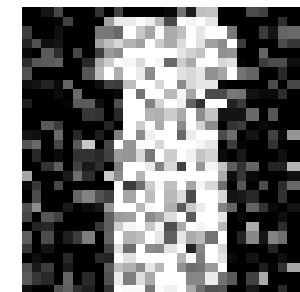
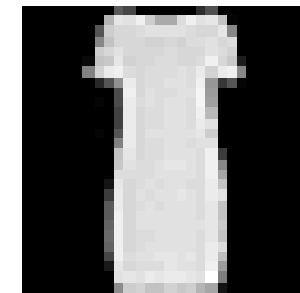
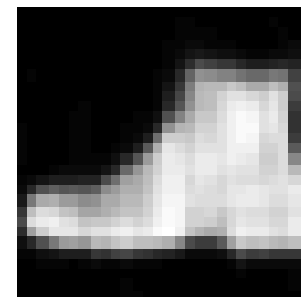
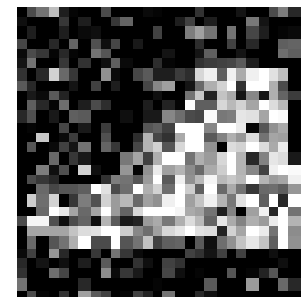
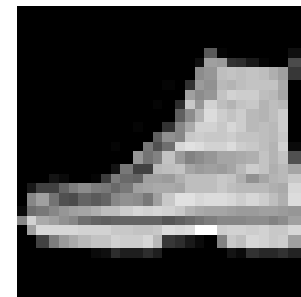
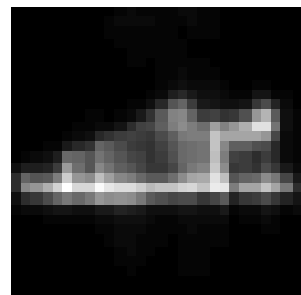
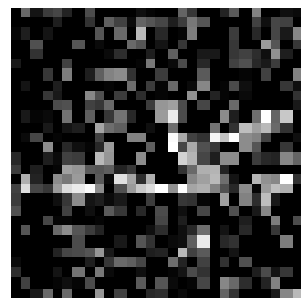
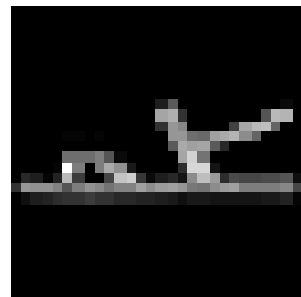
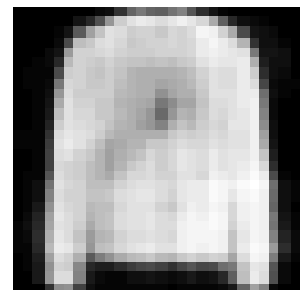
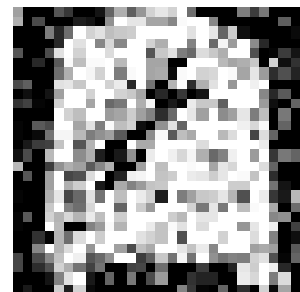
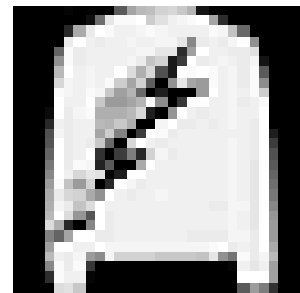


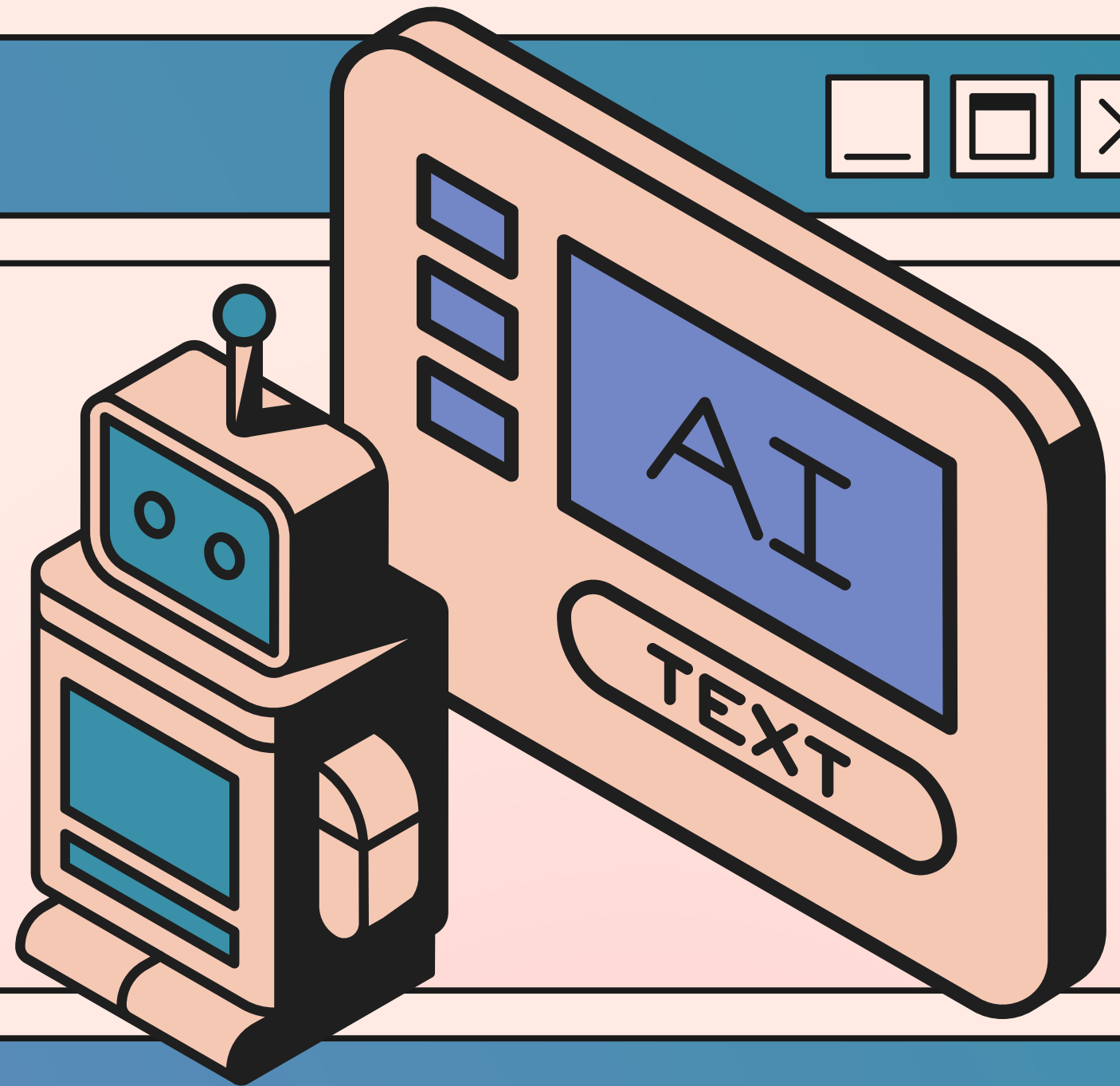
IMAGEM ORIGINAL x COM RUÍDO x RECONSTRUÍDA





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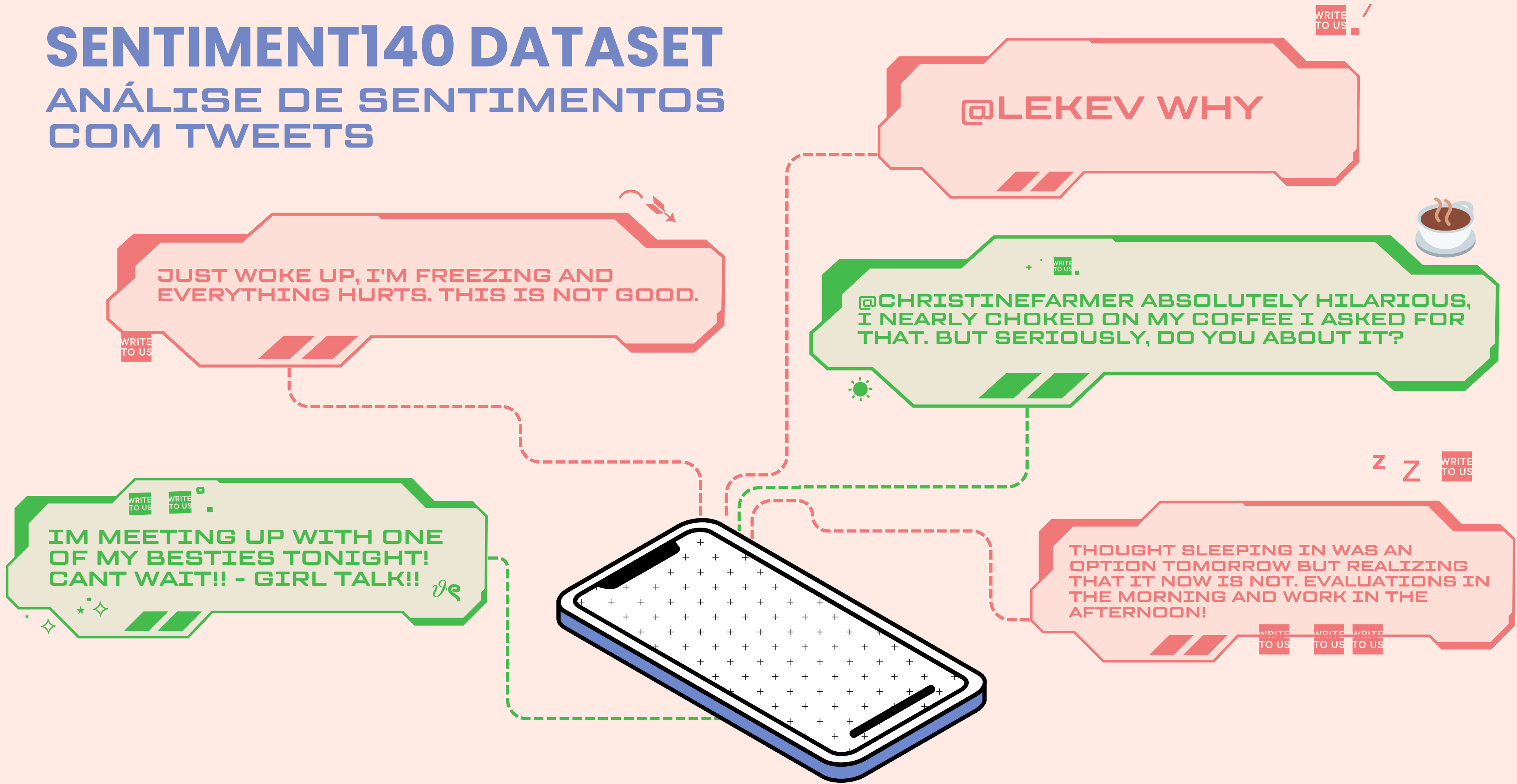
QUESTÃO 5



GISELE MENEZES , MARIANA MARTINS E THIAGO RODRIGUES

SENTIMENT140 DATASET

ANÁLISE DE SENTIMENTOS COM TWEETS



PRÉ-PROCESSAMENTO DOS DADOS

DF_ORIGINAL → DF_SMALL

- N = 50_000: número de amostras desejado.
- Mantendo a mesma proporção de classes que o dataset original.
- Binário: **0=neg, 1=pos.**

DF_CLEAN

- Substitui links → <URL>
- Menções → <USER>
- Hashtags → palavra
- Pontuação → <EXC>, <Q>
- Emoticons → <SMILE>, <SAD>, <LAUGH>
- Normaliza alongamentos (soooo → soo)

DF_STEMMED

- “Corta” palavras até sua raiz, sem se preocupar se vira uma palavra válida.
- **amando, amarei, amador → “am”**

DF_LEMMATIZED

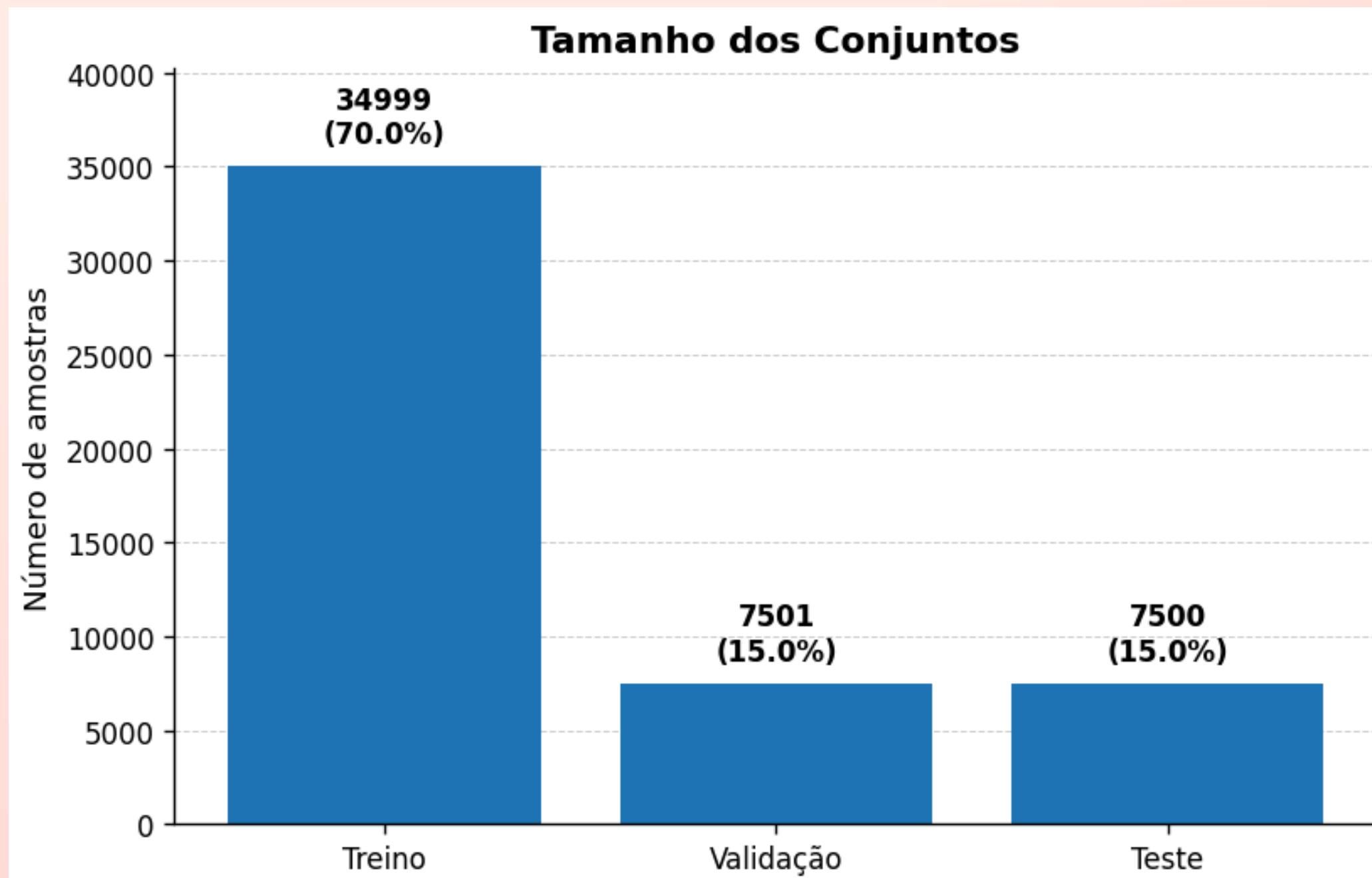
- Reduz palavras à forma canônica (lema), sempre uma palavra real do idioma.
- Considera classe gramatical e contexto.
- **amando, amarei, amador → “amar”**

PRÉ-PROCESSAMENTO DOS DADOS

target	text	clean	stemmed	lemmatized
0	Tired. Sleepy. Exhausted. Need my beauty sleep!	tired sleepy exhausted need my beauty sleep <exc>	tire sleepi exhaust need my beauti sleep <exc>	tire sleepy exhaust need my beauty sleep exc
1	@Net I wouldn't bet on it...	<user> i wouldn t bet on it	<user> i wouldn t bet on it	user i wouldn t bet on it

QUESTÃO 3

CONJUNTO DE VALIDAÇÃO



PIPELINE

& FLUXO

Pré-Processamento de Dados

build_vocab_from_texts

criar o vocabulário que converte palavras em números

<EXC>:1, tired:2, sleepy:3, ...

sequences_to_tensor

transforma todas as sequências em tensores fixos.

[2, 3, 4, 5, 6, 7, 8, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[B=1, T=26]

make_data_loader_from_texts

Divide os dados em batches para cada variante

encode_text (por texto)

o texto vira uma sequência de índices

[2, 3, 4, 5, 6, 7, 8, 1]

stratified_idx

divide os dados em train/val/test, preservando a proporção de classes

TEXTOS, RÓTULOS y

tired sleepy exhausted
need my beauty sleep
<EXC>

train_one

epochs = 7, lr = $5e-4$, weight_decay = $5e-4$,
loss: **BCEWithLogitsLoss(pos_weight)** - entropia cruzada binária

Entrada:
[B,T]

Embedding (pad_idx) + Dropout

Cada índice é substituído por um
vetor de dimensão 300
emb_dropout=0.2

2 → [0.12, -0.08, 0.34, ..., 0.05]
3 → [-0.22, 0.09, 0.14, ..., -0.18]
0 → [...]
...

[B=1, T=26, emb_dim=300]

Packing com lengths

as sequências são
compactadas considerando
seus comprimentos reais.

GRU

input_size = 300 (emb_dim)
hidden_size = 128 (HIDDEN)
num_layers = 1
bidirectional = True

Saída: (2, B, 128)
concatena direções → (B, 256)
[Juntamos os dois vetores (frente e trás)]

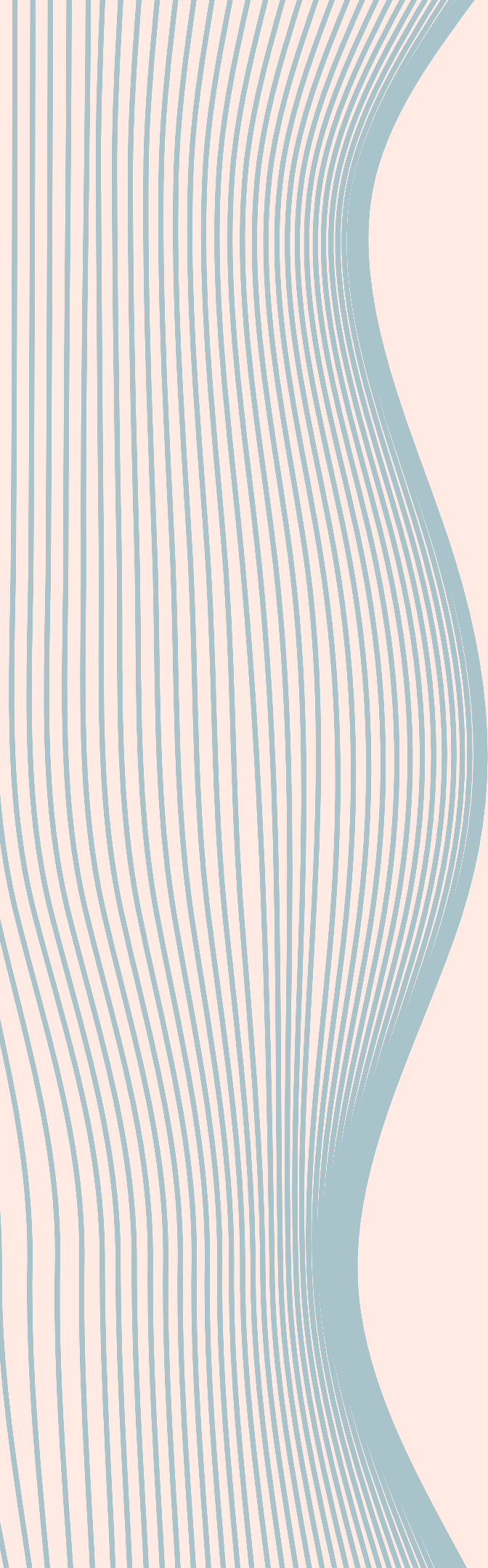
Classificação

Dropout(0.5)
Linear(256 → 1)

Saída logits: (B,)

Validação (eval_on)

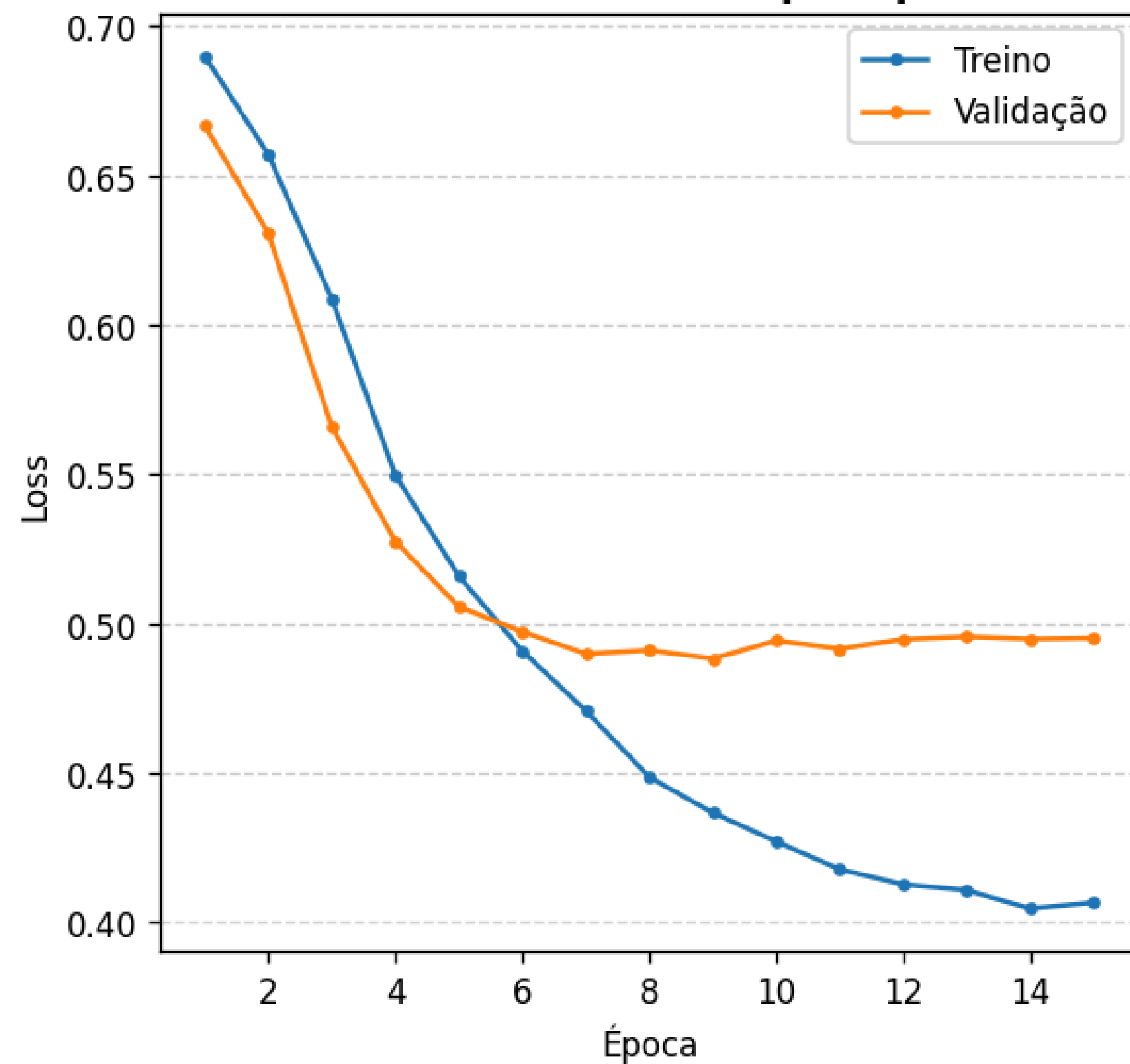
acc, y_true, y_prob(sigmoid-
probabilidade entre 0 e 1.)



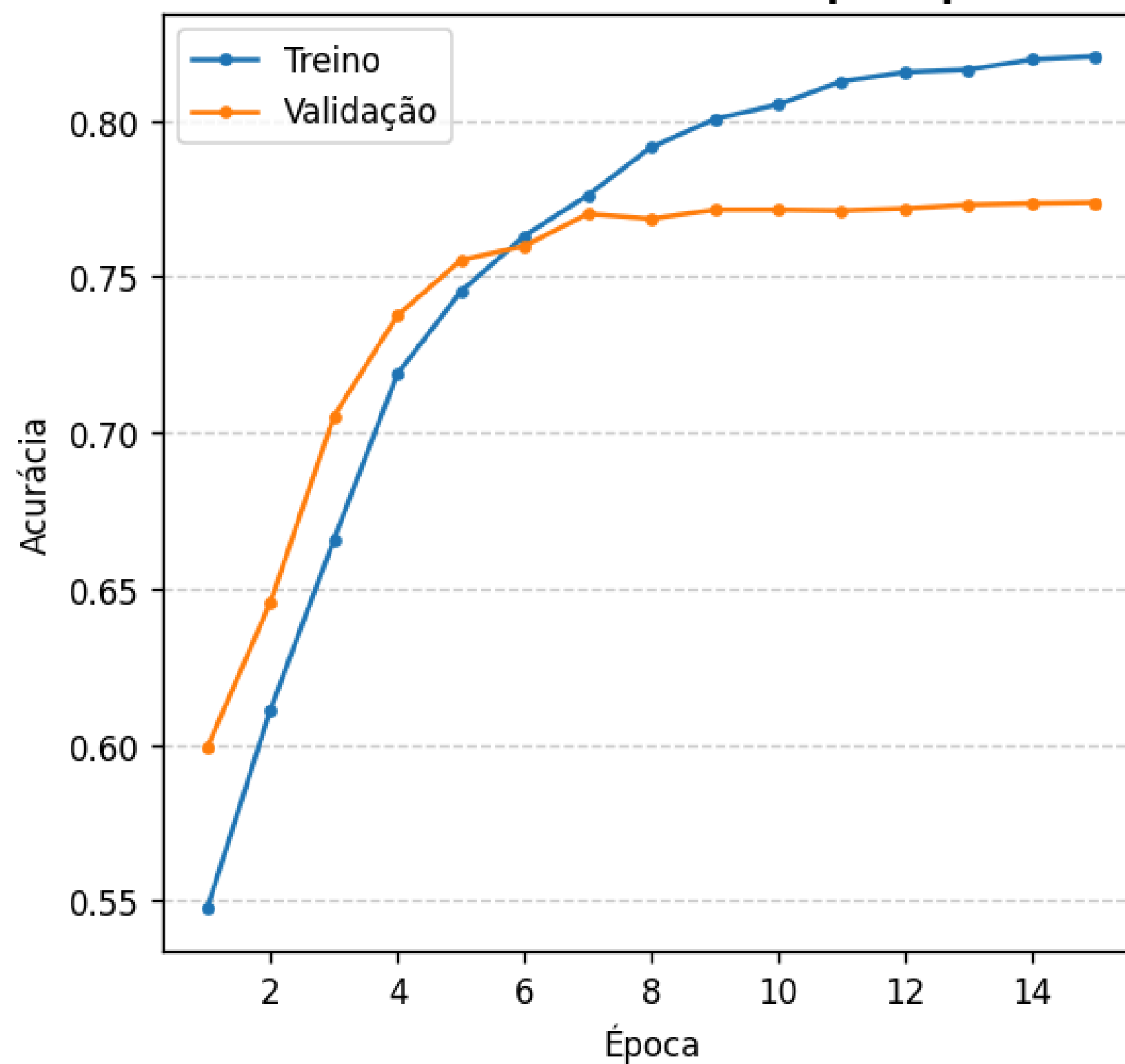
RESULTADOS

VARIANTE	ACURÁCIA (TESTE)
lemmatized	0.778
stemmed	0.771
text	0.769
clean	0.767

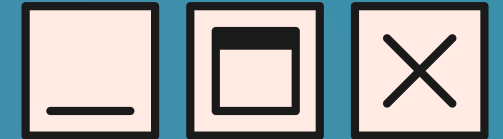
SentimentGRU - Loss por Época



SentimentGRU - Acurácia por Época



EXEMPLOS DE CLASSIFICAÇÃO



	Variante	Texto	Rótulo_verdadeiro	Predição	Probabilidade
0	lemmatized	no 8th win for rossi very sad	0	0 ✓	0.0156
1	lemmatized	user ahh you tweet ecstatic com lt 33 exc suck...	0	0 ✓	0.2157
2	lemmatized	go to bed let hope exc have trouble sleep thin...	0	0 ✓	0.0563
3	lemmatized	user sonny with a chance be so cool i hope you...	0	1 ✗	0.9278
4	lemmatized	user i don t have any of those color of course...	0	0 ✓	0.2628



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LISTA DE EXERCÍCIOS



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